Computerized Payroll System for Home of Job Enterprises (HOJE)

A Project

Presented to

the Faculty of STI College Calamba

In Partial Fulfillment

of the Requirements for the Degree of

Bachelor of Science in Information Technology

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March 2016

ADVISER'S RECOMMENDATION SHEET

This Project entitled

Computerized Payroll System for Home of Job Enterprises

(HOJE)

by :

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submitted in partial fulfillment of the requirements of the
Bachelor of Science in Information Technology degree
has been examined and is hereby recommended
for acceptance and approval

_____

Jerald A. Sanchez
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March 2016

PANEL'S APPROVAL SHEET

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Computerized Payroll System for Home of Job Enterprise (HOJE)

After having been presented is hereby approval

by the following members of the panel

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March 2016

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This Project entitled

Computerized Payroll System for Home of Job Enterprises

(HOJE)

After having been recommended and approved is hereby accepted
by the Information Technology Department
of STI College Calamba

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March 2016

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Panel's Approval Sheet

Project Coordinator and Dean's Acceptance Sheet

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Abstract

Payroll is defined as method of administrating employees' salaries in the organizations. The process consists of calculation of salaries and tax deductions of the employees, administrating the retirement benefits and disbursement of salaries to employees. It can also be called as an accounts activity which undertakes the salary administration of employees in the organization. Administrating the employees' salaries is not an easy task, the HR and accounts department work together to calculate and disburse the salary to the employees.

The specific tasks involved in payroll administration tend to vary according to the needs of each unique company or organization. For example, some companies have workers that receive commissions in addition to salaries. In these companies, including commissions in employee salaries is a part of payroll administrations. For some companies, it also involves assigning exempt or non-exempt status to workers, adding bonuses into pay checks, calculating overtime Payment's, and adhering to applicable employment law.

Handling payroll-related problems are part of payroll administrations as well. If an employee is paid an incorrect amount or a direct deposit fails to go through, both things are payroll administration issues, likewise tax filing and deduction errors are payroll administration matter as well.

Preface

This documentation is the developer's IT special project at STI College Calamba entitled "Computerized Payroll System for Home of Job Enterprise (HOJE)". HOJE has a growing up employment, and there have many Agency branches around Laguna. They are having a problem in payroll computation processing, and creating payroll reports because the payroll staffs spend too much work in manually creating employee's time sheet summary before it computes payroll, and then, they will create manually payroll reports. That is why the developers come up with an idea of developing a software program to help HOJE, to ease manually payroll processing.

This documentation focuses on how to improve the current HOJE payroll system. The documentation contains the problem of the client, and how the developers find a solution on how to solve the problem. Instead of manually creating payroll computation, the developers build software program that could compute automatically time keeping before it goes in payroll computation, by reading Excel Time Sheet Data, which first converted in Biometrics, but it is also depends in companies' client that has a Biometrics. This documentation is trying to shown readers how the developers solve the existing problem of the client by making computerized payroll system.

Chapter 1 INTRODUCTION

The world is rapidly changing, pushed by extreme developments in information and communication sciences and technology. The emergence of new technologies and the importance of software innovation have become driving forces of business. Computers changed the forms of all business aspects. They are sweeping away old ways of doing things with their speed and reliability. Computerized systems are version of manual organization systems dedicated in handling the organizations transaction, many industries, companies and even institution discovering applications of computerized system.

Payroll system largely came about during the industrial revolution of the 19th century. With the onset of mass employment and production arose the need to keep accurate records of workers and their pay. This is when the term "payroll" was first used. Early IT-based payroll system began to take shape in the 1950′s when the LEO (Lyons Electronic Office) was developed in order to handle clerical tasks. Whilst valuations, it was soon used to manage other clerical tasks including payroll services (CPS - Corporate Payroll Solutions Ltd., 2013). From that, other IT-based payroll systems are developed.

Hoje Job Contractor And Allied Services a single proprietorship company, composed and manage by different kinds of professionals. Hoje means Home Of Job Enterprises; Hoje Job Contractor and Allied Services is a private recruitment and placement agency situated at S-8 Teodora Business Center II Paseo de Alcasid, Crossing, Calamba City Laguna.

Hoje Job Contractor And Allied Services and Home of job enterprises is generally managed by Mr. Davebert P. Honculada, A graduate of Information Technology of Father Saturnino Urios

University, Butuan City, Agusan Del Norte and Civil Service Passer. Home of job enterprises is located at S-8 Teodora Business Center II Paseo de Alcasid, Crossing, Calamba City Laguna.

1.1 Statement of the Problem

1.1.1 General problem

Home of Job Enterprises (HOJE) is facing some issues in their manual payroll system. Manual system makes their process time consuming, unsecured data of employees and data can be lost.

The study's concern is the development of the Computerized Payroll System, aiming to answer the problems encountered and to improve the current system of HOJE, particularly to the fields concerned Faster calculating of employee's salary, secure payroll records and generate reports.

1.1.2 Specific Problem

- **How to create and design a system that will hasten payroll system?**

  Encoding and computing of employees total working hours and salary can be time consuming since it is does manually.

- **How to create and design a system that will generate reports?**

  The developers notice the manual works of creating payroll reports which is the payroll, government contributions, statement of accounts, summary of deductions, and billing computations summaries.

- **How to create and design systems that make data secured?**

  HOJE place some of their employee's data in a filing cabinet and others are placed in Excel file that has no password, resulting to its vulnerability to falsification of data.

- **How to create and design a system that provides storage that will prevent loss and damage of data?**

  Files are being transferred by company's coordinator to Hoje employee. Thus, this process is prone data loss and damage.

- **How the adapting to the new technologies available in the market like the computerized system technology and integrating it to the current system will satisfy the needs of Hoje in a strategic way?**

  Hoje is currently using manual way of computing the salary of employee.

1.2 Current State of the Technology

- **Client**

Home of Job Enterprises uses Microsoft Excel to compute the salaries of the employee. The payroll happened every cut off period. First, the coordinator gets the Time Sheet of the employee to analyze and check manually to create the Time Keeping Summary.

The company HR checks the Time Keeping summary, to turn over to the Accounting department in order to make billing payroll summary, which includes the total gross pay of the employee and employer's contributions, and checked by the Accounting Supervisor, reviewed and accepted by Office Administrator, and approved by the Owner of the company. Afterwards, the Coordinator prepared the Government contributions approved by the Owner of the company, to make the payroll summary which includes the total gross pay of the employees, and Government remittances.

Lastly, the Accounting staffs make Statement of Accounts which is the Reimbursable Salaries, and Government Contributions of the company.

- **Local**

Andrew Mendez (2007) a programmer, made a similar study for Vision Designer Inc. It has an identification machine aimed at verifying a person entering through the swipe identification cards. After entering their given code number, the identification cards swiped the machine cards reader trigger the verification. If access granted the micro switched swiped will then automatically unlock to allow and enter a person.

- **Foreign**

Columbia State Community (2006) uses computer identification account (Log in name Password) for access to the institutions computer usage logs occur electrically through the individual user login/out process. Users are also informed that they are responsible for any computer files, database and internet sites that are access their computer identification account, not withstanding their failure to adhere to the log in/out process, on their voluntary publication of their account information to other.

1.3 Objectives

1.3.1 General Objectives

The main objective of this study is to design and develop an Computerized Payroll System for Hoje that will reduced time consuming process of computing total working hours of employee in payroll, and to manage better manual way of computing payroll.

1.3.2 Specific Objectives

- **To create and design a system that will hasten payroll system.**

  The developers developed a computerized payroll system that will hasten their manual payroll system.

- **To create and design a system that will generate reports.**

  The developers provided a system that can generate real time reports.

- **To create and design systems that makes data secured.**

  Hoje place some of their employee's data in a filing cabinet and others are placed in excel file that has no password, resulting to its vulnerability to falsification.

- **To create a system that will enhance the security of data.**

  The developers used a database system which organizes the information of employees so that it can be easily access, manage, and update. The system creates back-up and restores data.

- **To adapt to the new technologies available in the market like computerized system technology and to integrate it to the current system of HOJE to satisfy their needs in a specific way.**

  To create and design a computerized payroll system thatwill integrate to the current technology of Hoje for fast and efficient payroll process.

1.3.3 Scopes and Limitations

1.3.3.1 Scopes

- **User Authentication Module–** This will be use by the accountant to access and operate the system. This module contains textbox which the accountant enters his/her username, a masked textbox for password field, and login button where if you clicked thee button it will validate the account through database. Username and Password are needed to be valid in order to use and access the system**.** This will serve as a security check to the user. It will require a valid username and password to access the main form. If invalid username and password as entered a warning message will appear.
- **User Account Management Module-** This form contains the settings where the use password and username can change password or username.
- **Employee Information Module –** The developers develop this module to help the client to add information of the new employees and can edit existing information.
- **Report Management Module –** To let the client view pay slip and payroll summaries including statement of account, deduction summaries, and government contributions.
- **Payroll Management –** The module could manage the gross pay, and other payroll deductions. The payroll deduction contains government deduction and company deductions.
- **Leave Schedule Management –** The developers develop a module that will let the client add and monitor the leave schedule of the employee. The module includes the employee information, number of days, and reasons of leave.

- **Time Keeping Module –**The module could import the excel time sheet files and create time sheet summary.
- **Payroll Settings Module –** The module has:
  - Computation settings could update new percentage of computations.
  - Time Standard settings could update the standard working schedules, lunch break settings, allowable tardiness and under time, and core hours standard.
  - Holiday Management settings which includes holiday events.
  - Deduction Table settings could update SSS, PHIC, and PAGIBIG tables.

1.3.3.2 Limitations

- The system unit must have a Wamp Server application to run the program.
- The loans are not issue to the payroll management of HOJE Company.
- The program did not issue Alpha list and other tax reports.
- The program could only read in military time.
- The program is dependent on right excel time sheet pattern that import in time keeping system.
- The employee's salary will not be sent through Automated teller machine (ATMs) or cheques.

Chapter 2 THEORETICAL FRAMEWORK

2.1 Introduction

A Payroll System must be user friendly interface, which determines how system can be used, and of causes reliable database management system for storing important data. This chapter discusses all the possible protocols, algorithms, and courses of action of needed to develop the whole study and also the theoretical that lead to design of the proposed system. Payroll system is an example of Transaction Processing System. According to Dr. Ghaeibi, Lecture of Institute for Sustainable Resources (ISR) Transaction Processing System or TPS is a computerized system that performs and records daily routine transactions necessary to the conduct of the business. TPS are information system that process data resulting from the occurrence of business transactions.

In developing a Payroll System the developers decided to use the following sets of existing technologies: MySQL Database management system for storage and retrieving of information's, NetBeans IDE 7.4 for coding and designing the user-interface.

2.2 Methodology of Study

Rapid Application Development (RAD)

According to James Martin, **Rapid Application Development (RAD)** is a development lifecycle designed to give much faster development and higher-quality results than those achieved with the traditional lifecycle. It is designed to take the maximum advantage of powerful development software that has evolved recently.

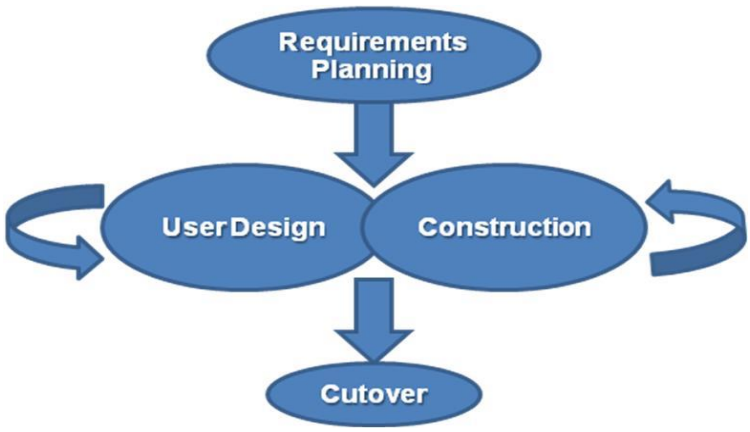


Figure 2-1 Rapid Application Development Methodologies

The James Martin approach to RAD divides the process into four distinct phases:

- **Requirements Planning Phase**

  The first phase in RAD is called Requirements/ planninghere the developers studied the system used by the client, which is the manual type. That system was evaluated, and the developers came up with the idea of creating a computerized payroll coded, and built the propose system using preferred programming language like Java Netbeans and MySQL database.

- **User Design Phase**

  The second phase in RAD is called System Design phase. Here, the developers will do the demonstration for every module of the proposed system to check whether these modules successfully met and achieved its functions.

- **Construction Phase**

  The third phase is Refining Construction Phase. Here, all the refinements and modification of built system will be implemented. The developers will focus on the modular approach of the system.

- **Cutover Phase**

  Here, the developers will check the proposed system for bugs or errors. Assuming that the system works fine, it will then be deployed to the client to undergo unit and functional testing.After satisfying the conditions of the testing phase, the developers will implement the system to the client.

2.3 Transaction Processing System

A payroll system is a type of Transaction Processing System. Transaction processing system for business transactions is involving the collecting, modification and retrieval of all transaction data. Transaction Processing in Payroll has different stages.

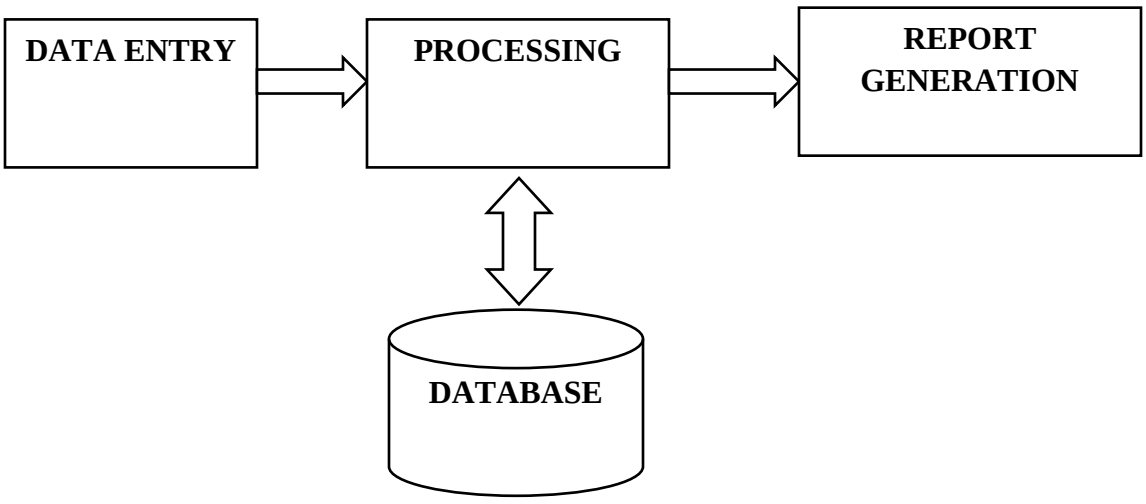


Figure 2-2 Transaction Processing Systems

- **Data Entry**

  In this stage the time sheet of the employee will be entered.

- **Processing**

  Next stage is the processing of data; here the system will compute the total salary of the employee.

- **Database storage**

  The third stage is where the process data will be recorded for future references.

- **Report generation**

  The final stage is report generation stage; all the reports about the processed data will print.

2.4 Database Management System

Database Management System is a complex set of software programs that controls the organization, storage, management, and retrieval of data in database. DBMS are categorized according to their data structures or types, sometime DBMS is also known as Data Base Manager. It is a set of pre written programs that are used to store, update and retrieve a database. It is a complex set of software program that controls the organization, storage, management, and retrieval of data in a database.

2.4.1 MySQL

MySQL is a freely available open source Relational Database Management System (RDBMS) that uses Structured Query Language (SQL). SQL is the most popular language for adding, accessing and managing content in a database. It is most noted for its quick processing,

proven reliability, ease and flexibility of use. MySQL is an essential part of almost every open source PHP application.



Figure 2-3 MySQL

2.5 NetBeans

NetBeans IDE 7.4 is an open-source integrated development environment. NetBeans IDE supports development of all Java application types. The NetBeans platforms allow applications to be developed from a set of modular software components called modules. The NetBeans is primarily intended for development in java, but also supports other languages, in particular PHP, C/C++ and HTML.

NetBeans is a powerful Graphical User Interface builder (GUI) and it offers Rapid Application development for increased developers productivity. It enables compilation and execution all in one step. NetBeans also have build in Java Database Connectivity that will allow the developers to easily connect the system with the database.



Figure 2-4 NetBeans IDE

2.6 Summary

The use of those theories and proper resources will make the system functional and easy to use. By using Rapid Application Development Method the developers identified the steps for creating the systems. The developers use NetBeans IDE 7.4 for the user interface and coding, and MySQL 5.6.1 for the database. As discussed earlier in the chapter. NetBeans IDE 7.4 is a programming language and integrated environment, so it is considered as a useful and relatively easy programming language for the beginner to learn Java Programming. Meanwhile, MySQL 5.6.1 is a relational database management system from Oracle Corporation. We can easily track and report information with ease with pre–built application phpMyAdmin.

The descriptions given above gives the user an assurance that the system would be powerful but user-friendly.

Chapter 3 COMPUTERIZED PAYROLL SYSTEM FOR HOME OF JOB ENTERPRISES (HOJE)

3.1 Introduction

In developing a system, developers are supposed to use different diagrams to represent the flow and scope of the system. This will serve as a guide of the developers of the system.

In this chapter, the diagram will be discussed. Diagrams have been used in order to simplify the understanding of the system. The Hierarchical Input Process Output (HIPO) diagram, the level of the system and input and output process will be shown. The entity Relationship Diagram (ERD) shows the relationship of the different data, usually from the database system. The Data Flow Diagram (DFD) includes different levels of the process according to the Hierarchy Diagram, the overall process of the system from the input to output. If the process and diagrams will be used efficiently and explained effectively, then the system by the developers will be understood easily.

3.2 Context Diagram

A Context Diagram represents a view of all system boundary of interest. A context diagram defines the system domain that is under investigation within an organization environment. It is used for establishing the boundary of the business or system under investigation. It defines the external entities along with major data interfaces that interact with the target process.

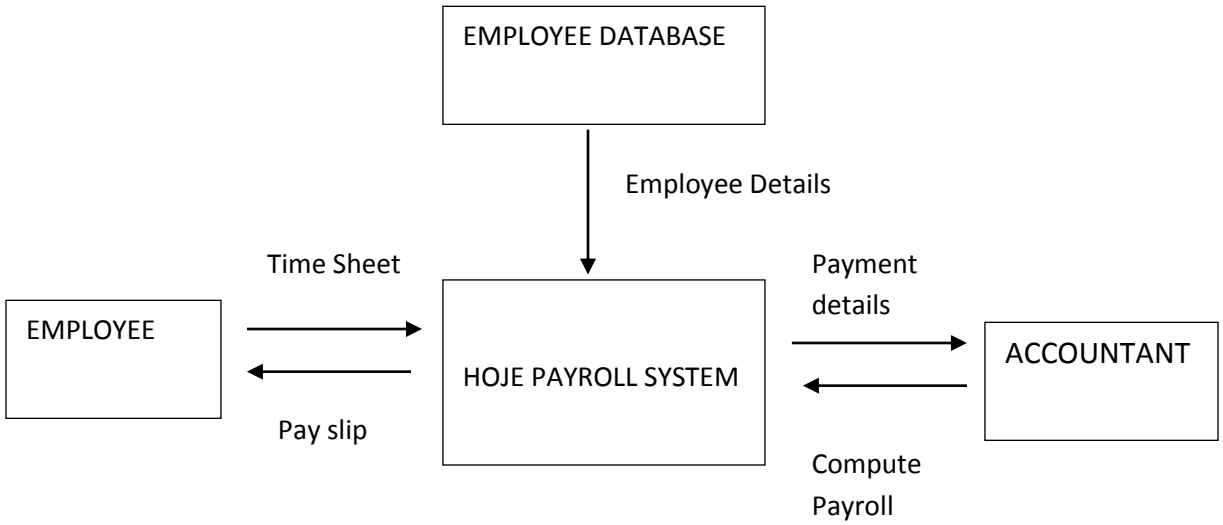


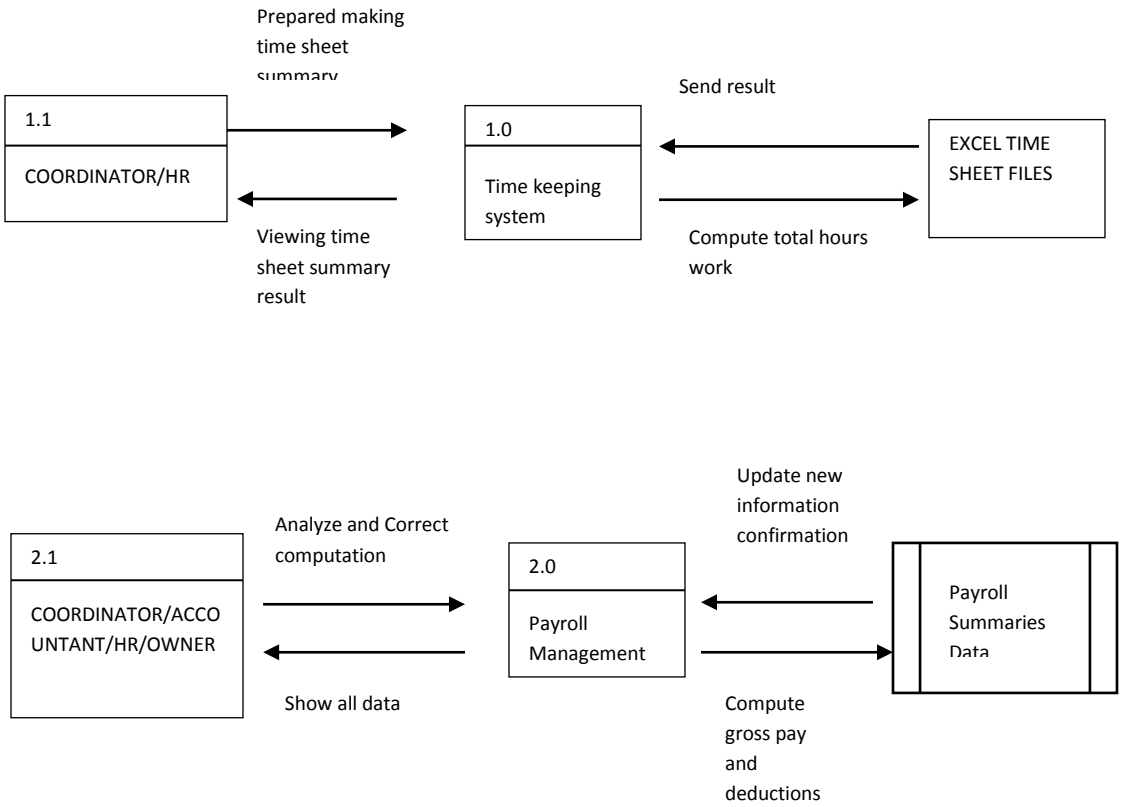
Figure 3-1 Context Diagram

The figure shown above is the context diagram. It shows a brief description of how the system runs. The accountant will input the number hours worked by the employee into the payroll system. The system will give the payroll report to the accountant and the accountant will print the pay slip to be given to the employee.

3.3 Data Flow Diagram

The Data Flow Diagram (DFD) is graphical representation of the flow of data through an information system but also as a way to model whole organizations. You build a DFD at the very beginning of your business process modeling in order to model the functions your system has to carry out and the interaction between to functions together with to focusing on data exchanges between processes. You can associate data with conceptual, logical and physical data models and object oriented models.

DFD LEVEL 0: HOJE PAYROLL SYSTEM



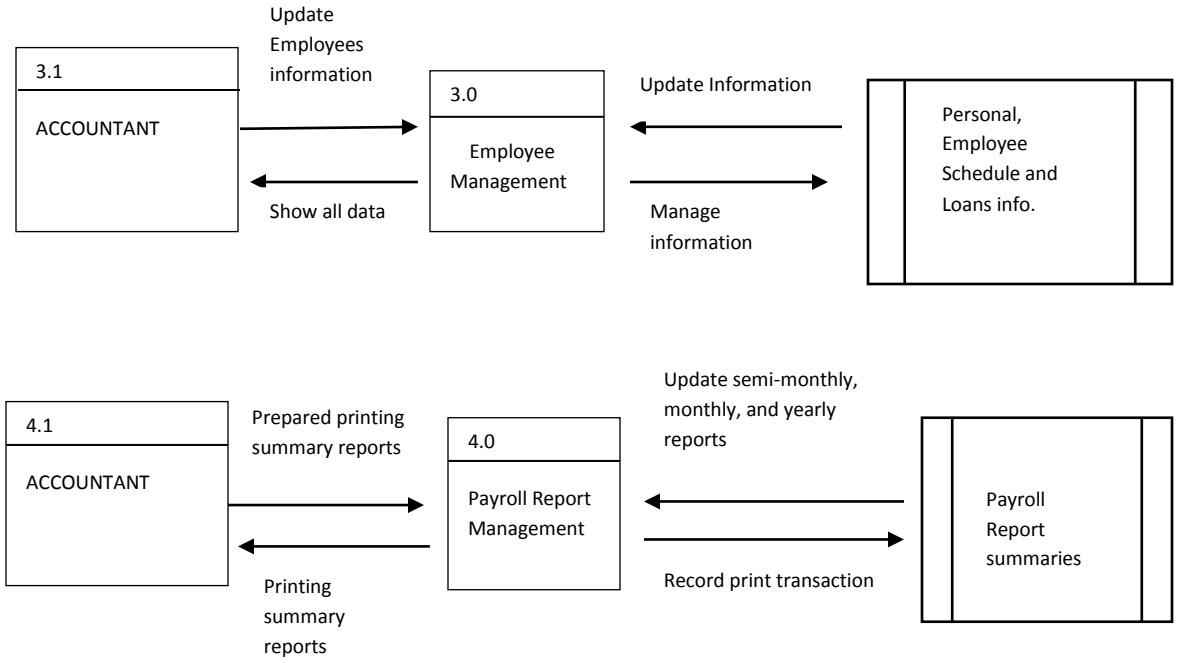


Figure 3- 2 Data Flow Diagram (Level 0)

The figure shows the inter-relationships between the components of the Computerized Payroll System for Home of Job Enterprises (HOJE). The Data Flow will starts when the accountant log-on to the system. It was then followed by any function to be used by the accountant. If the user wants to add of a new employee, the user must go employee information of an employee. The user can also add information of an employee it will saved to the database.

The user can input the timesheet of an employee and the total working hours will be computed. The salary will then deducted with beneficial deductible saved in the database and the information will be received by the user. From the database the user can generate reports like printing pay slip of an employee and payroll summary.

TIME KEEPING MANAGEMENT MODULE

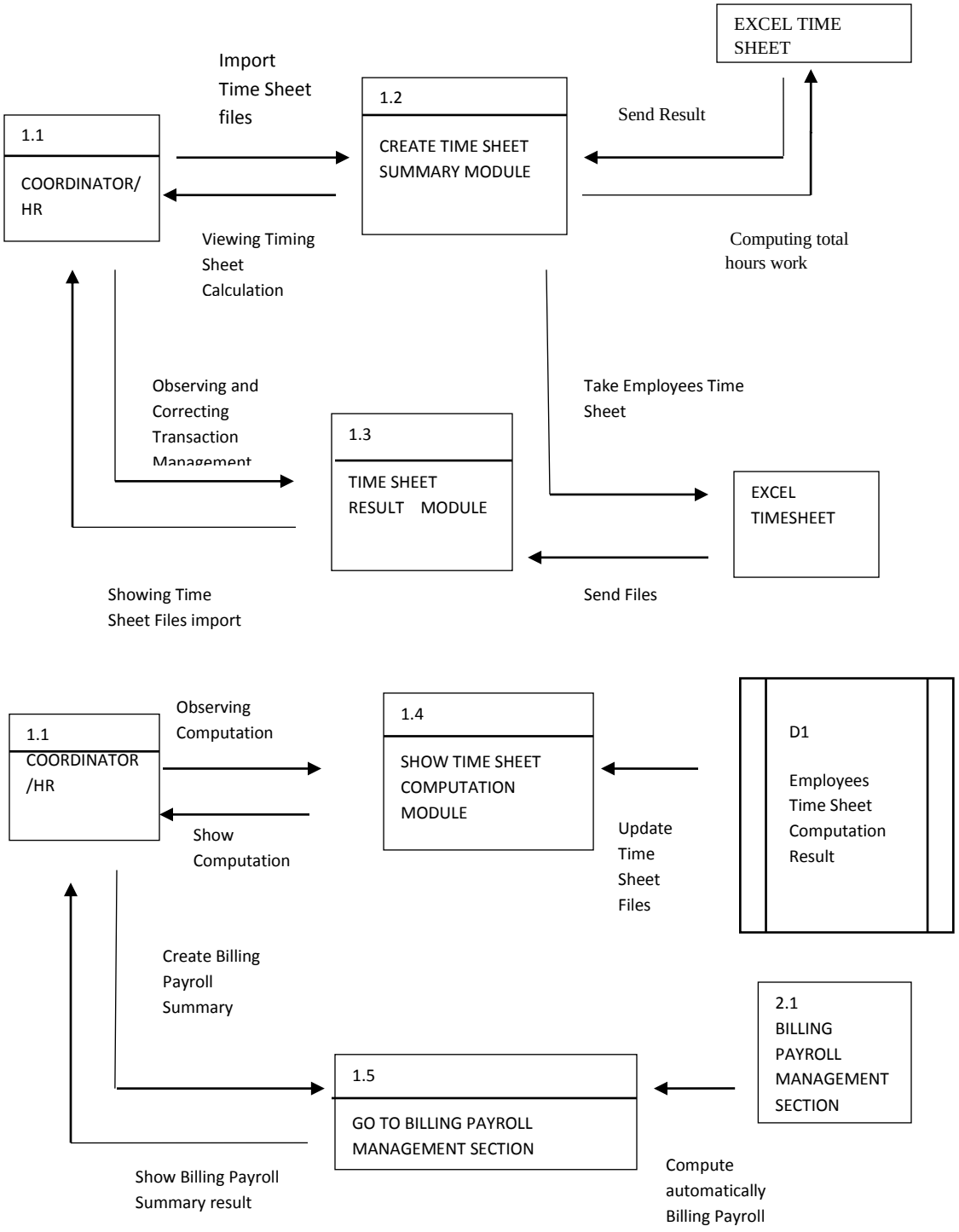


Figure 3-3 Data Flow Diagram (Level 1)

Figure 3-3 shows how the payroll staffs create time sheet summary by importing Excel time sheet data that is made by Biometrics. If they like to show how the program compute time keeping summary, Payroll Staffs must go in Time Sheet Result Module.

TIME SHEET RESULT MODULE

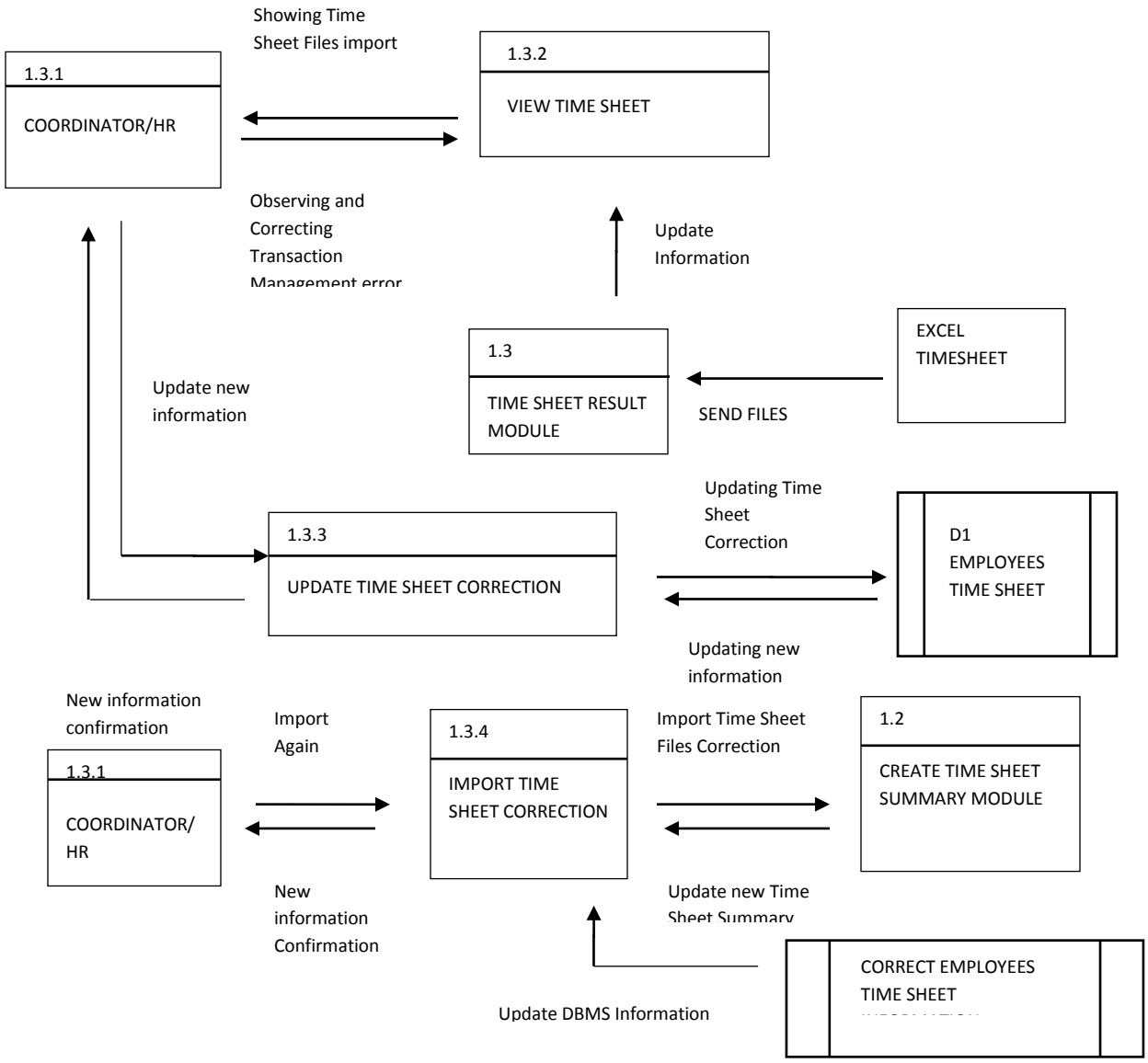


Figure 3-4 Data Flow Diagram (Level 1)

The figure shown above is called Time Sheet Result Module. The Payroll Staff show how the program compute time keeping summary by observing Employee's working time schedule to match up in overtime, regular day, legal holiday and rest day working hours of employees. This module could update another transaction if the Excel Time Sheet Data is corrupt by choosing data would update, and then clicking update time sheet correction button, and if Admin wants to change time sheet summary to compute again, Admin must choose employees and its data would update, and clicking import again button.

BILLING PAYROLL MANAGEMENT

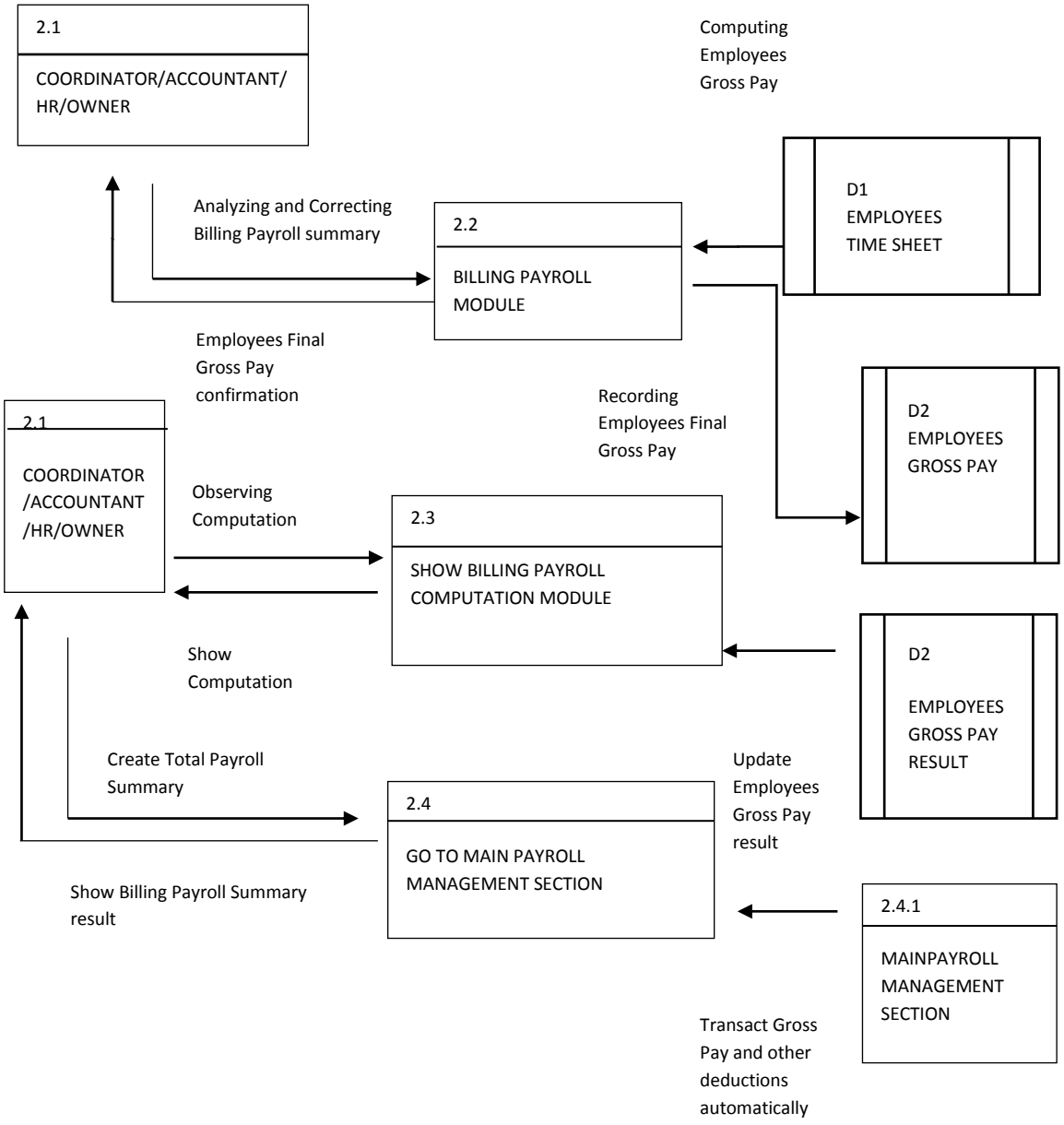


Figure 3-5 Data Flow Diagram (Level 2)

The figure shown above is called Billing Payroll Module. This shows how the Payroll Staff observe the program how employees gross pay compute the overtime, legal holiday, regular day, rest day, Admin fee to the employee, E-COLA, and Socio-Economic Allowance, minus to the total tardiness and under time of employee.

MAIN PAYROLL MANAGEMENT SECTION

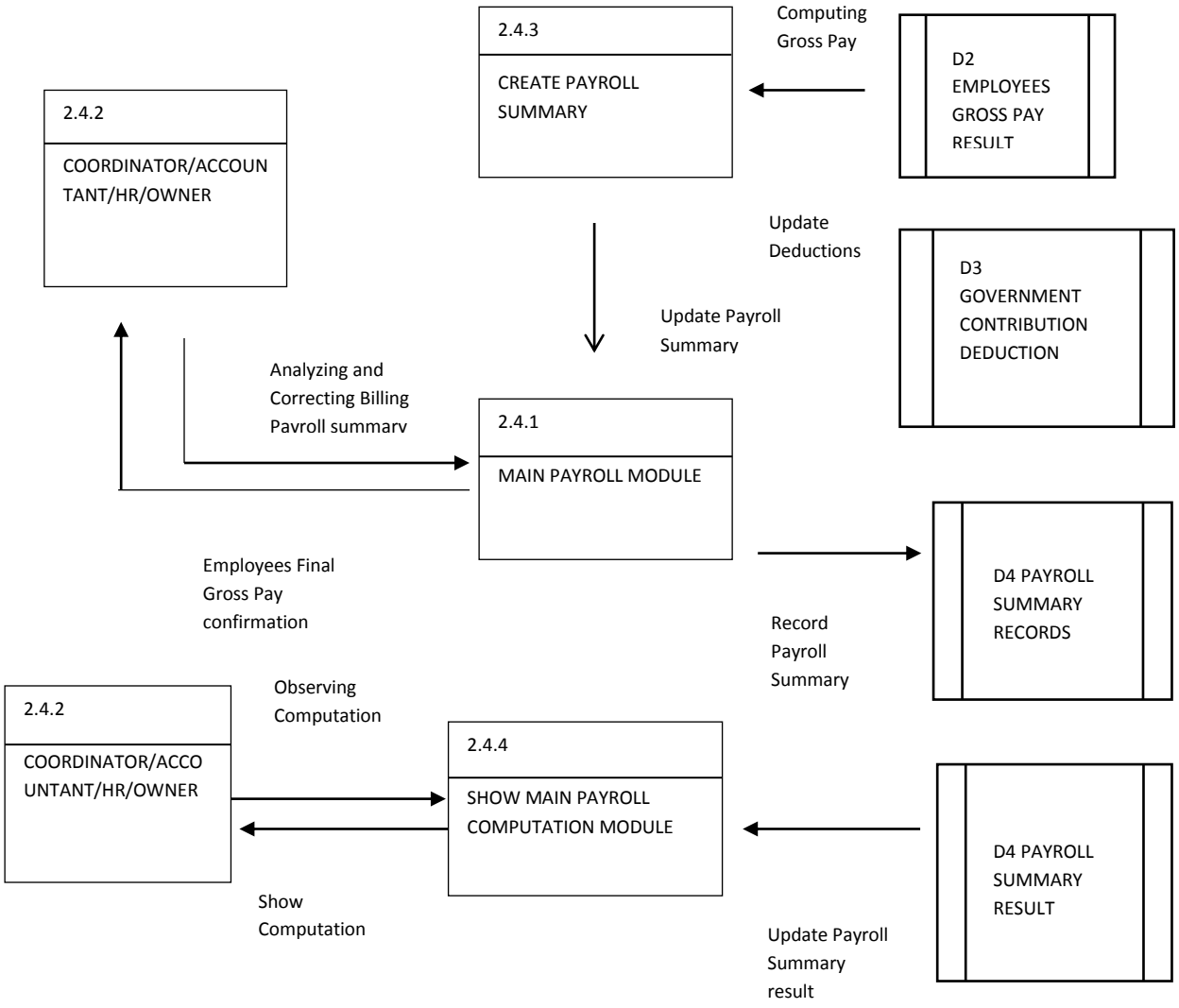


Figure 3-6 Data Flow Diagram (Level 2)

The figure shown above is called Main Payroll Module. This shows how the Payroll Staffs observe the program how employees total gross pay in billing payroll summary minus to the government deduction in SSS, Philhealth and PagIbig, plus in the medical, canteen and uniform deduction.

EMPLOYEE MANAGEMENT

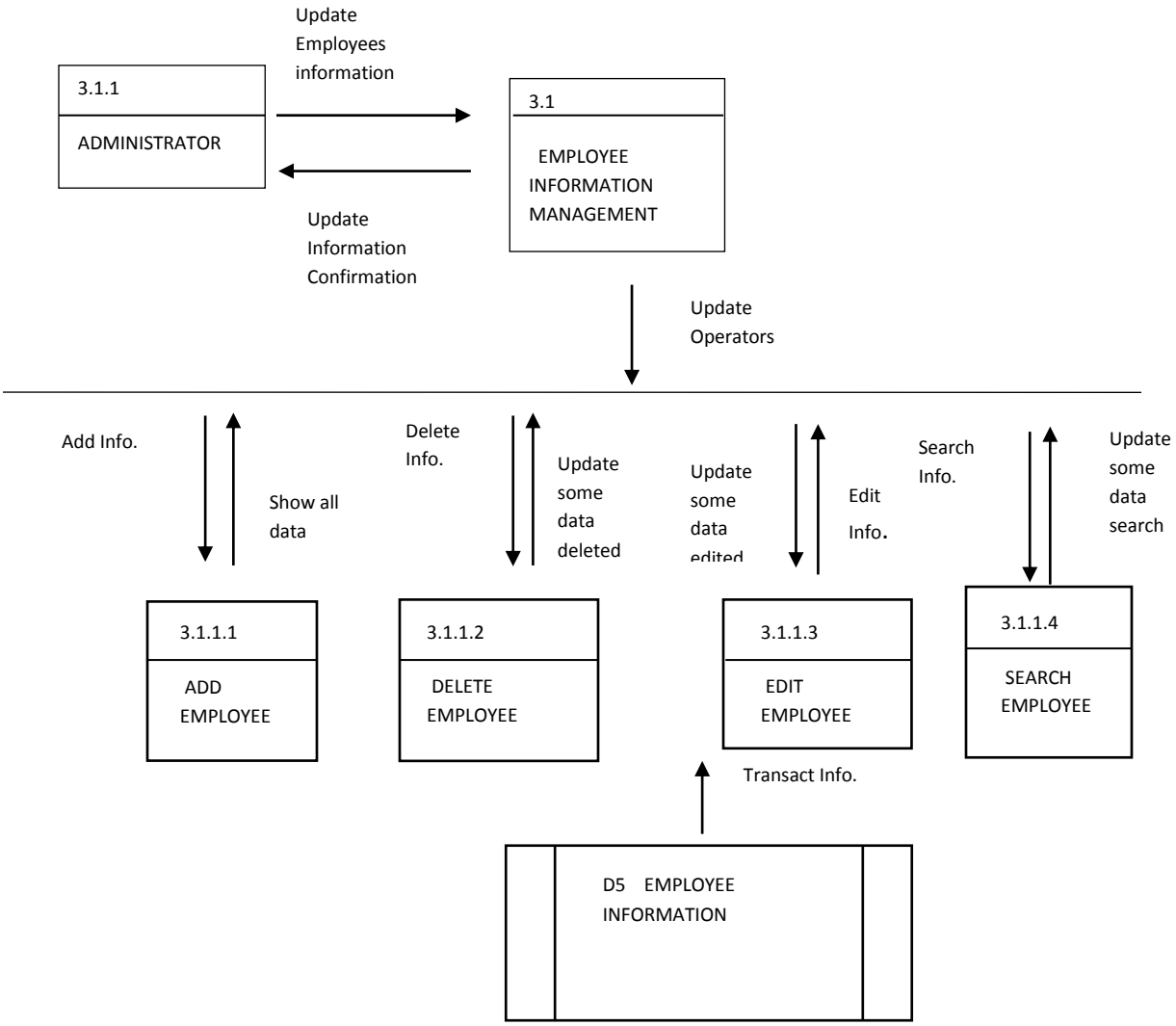


Figure 3-7 Data Flow Diagram (Level 3)

The figure shown above is called Employee Information Management Module. It shown the Administrator could only add, edits, delete and search employee's details in program.

JOB INFORMATION MANAGEMENT

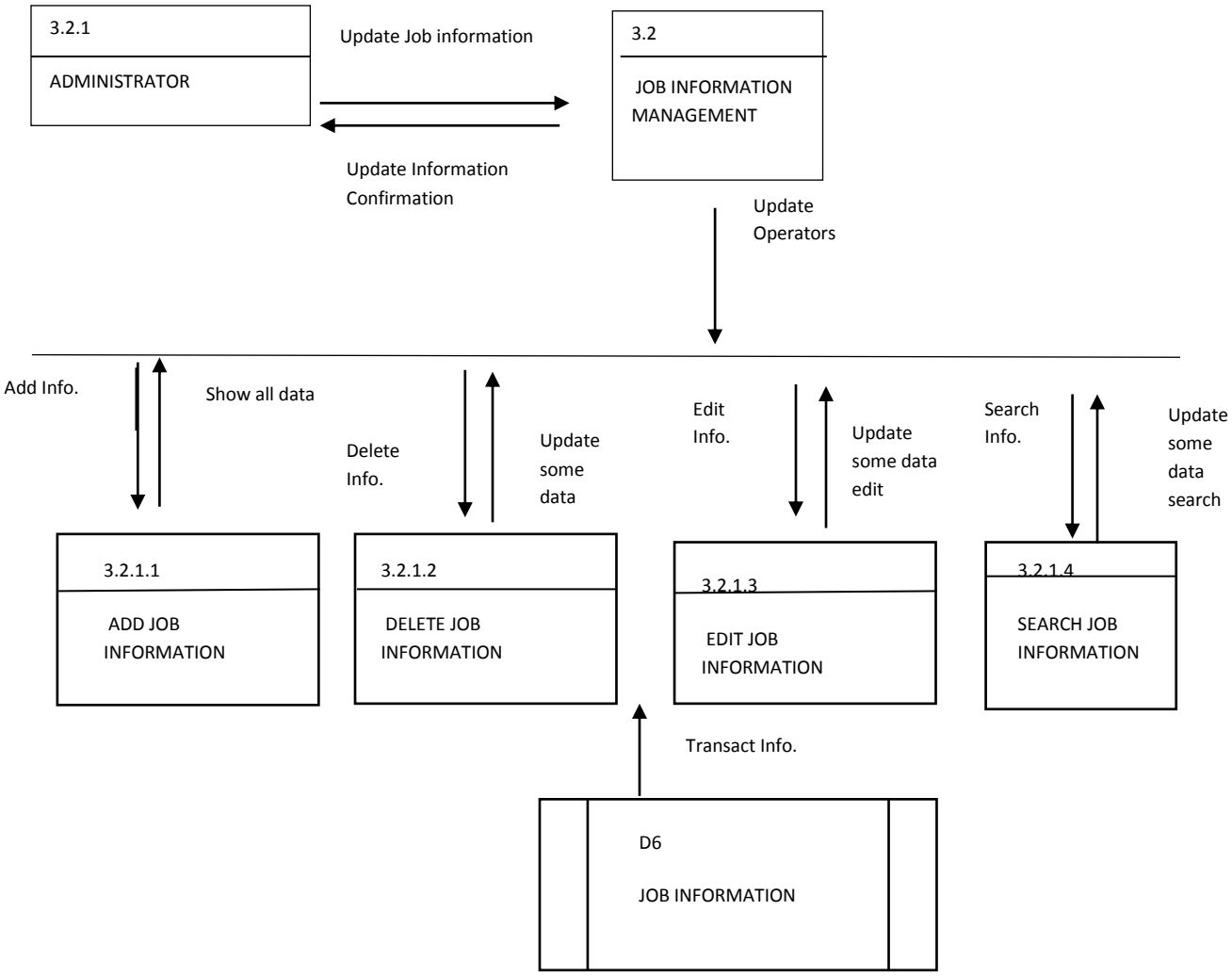


Figure 3-8 Data Flow Diagram (Level 3)

The figure shown above is called Job Information Management Module. It shown the Administrator could only add, edit, delete and search job details in program.

LEAVE SCHEDULE MANAGEMENT

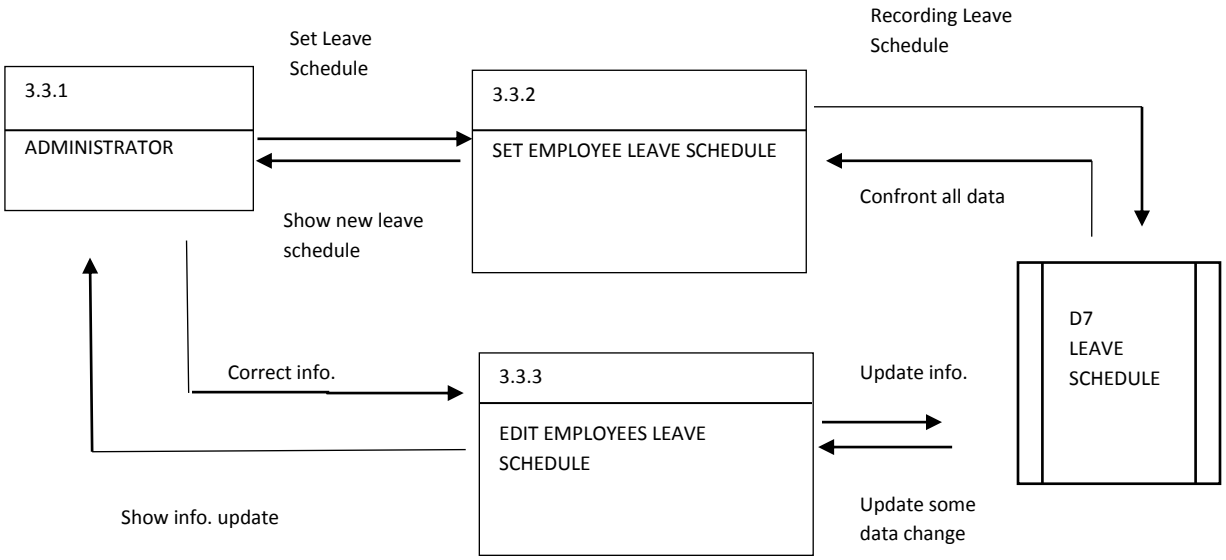


Figure 3-9 Data Flow Diagram (Level 3)

The figure shown above is called Leave Schedule Management. It shown the Administrator could only set leave schedule for the employees would leave in job.

EMPLOYEE LOAN MANAGEMENT

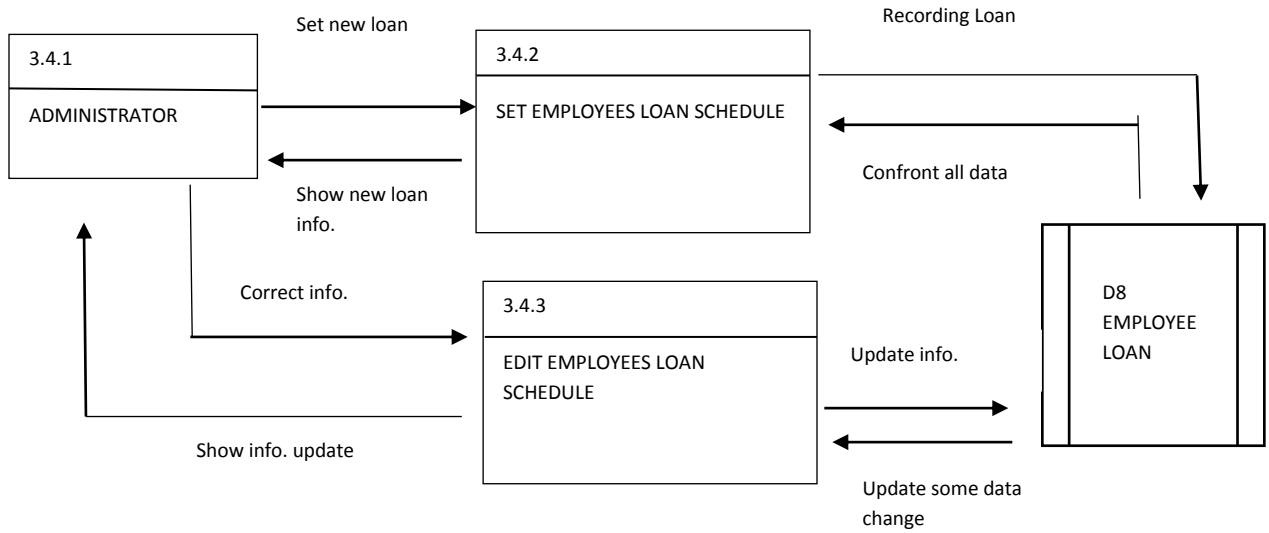


Figure 3-10 Data Flow Diagram (Level 3)

Figure 3-10 shown the Administrator could only update employee's loans.

EMPLOYEE BASIC RATE MANAGEMENT

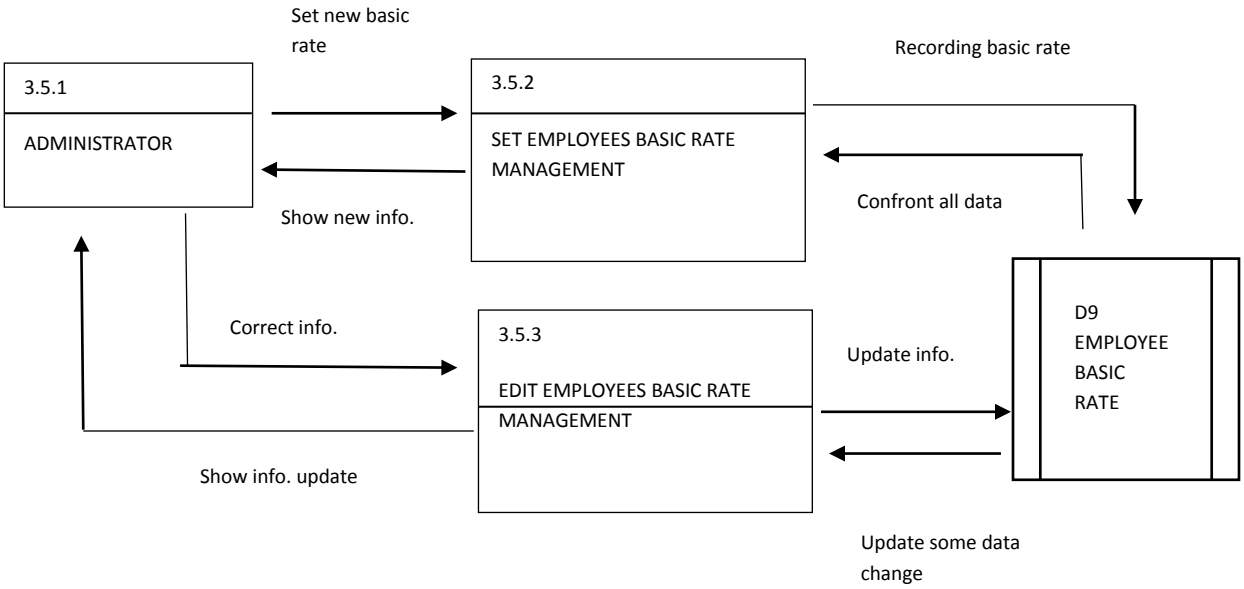


Figure 3-11 Data Flow Diagram (Level 3)

The figure shown above is called Employee Basic Rate Management Module. It shown the Administrator could only update employee's basic salary rate.

EMPLOYEES CUT OFF PERIOD MANAGEMENT

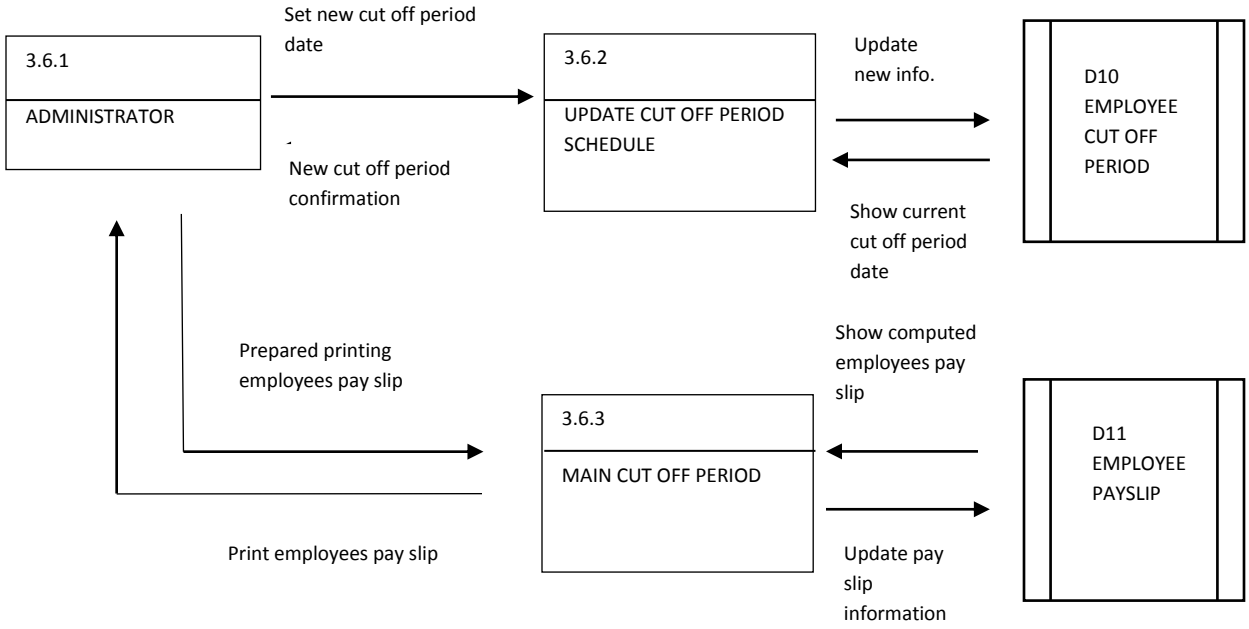


Figure 3-12 Data Flow Diagram (Level 3)

The figure shown above is called Employee Cut Off Period Management Module. It shown the Administrator could only update employee's payment schedule or cut off period.

PAYROLL REPORT MANAGEMENT

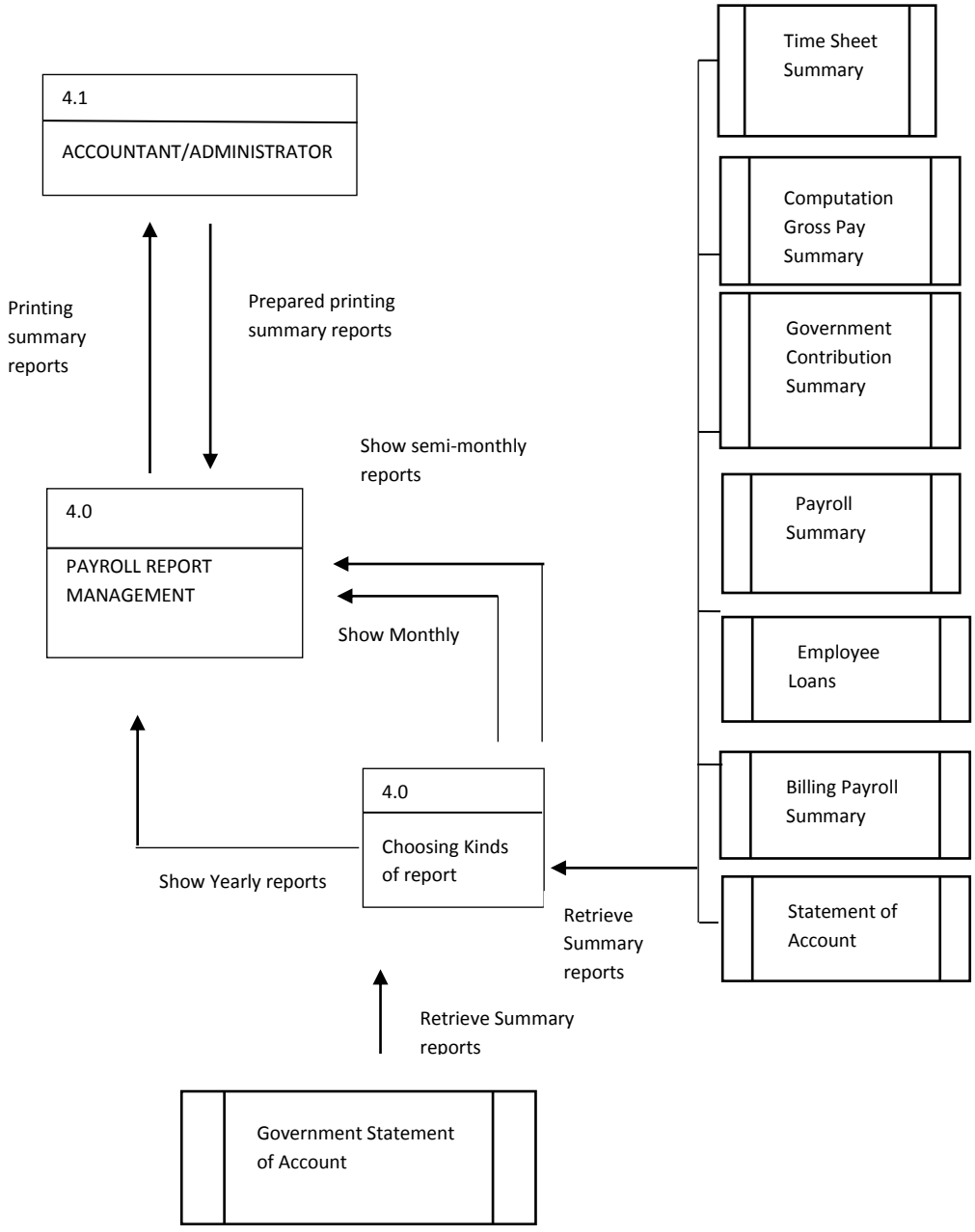


Figure 3-13 Data Flow Diagram (Level 3)

The figure shown above is called Report Management Module. It shown the Administrator or Accountant could only print payroll reports.

3.4 Hierarchal Input Process Output

Hierarchal Input Process Output is a system analysis design aid and documentation technique for representing the modules of a system as a hierarchy. This explains how the forms of modules are arranged inside the system.

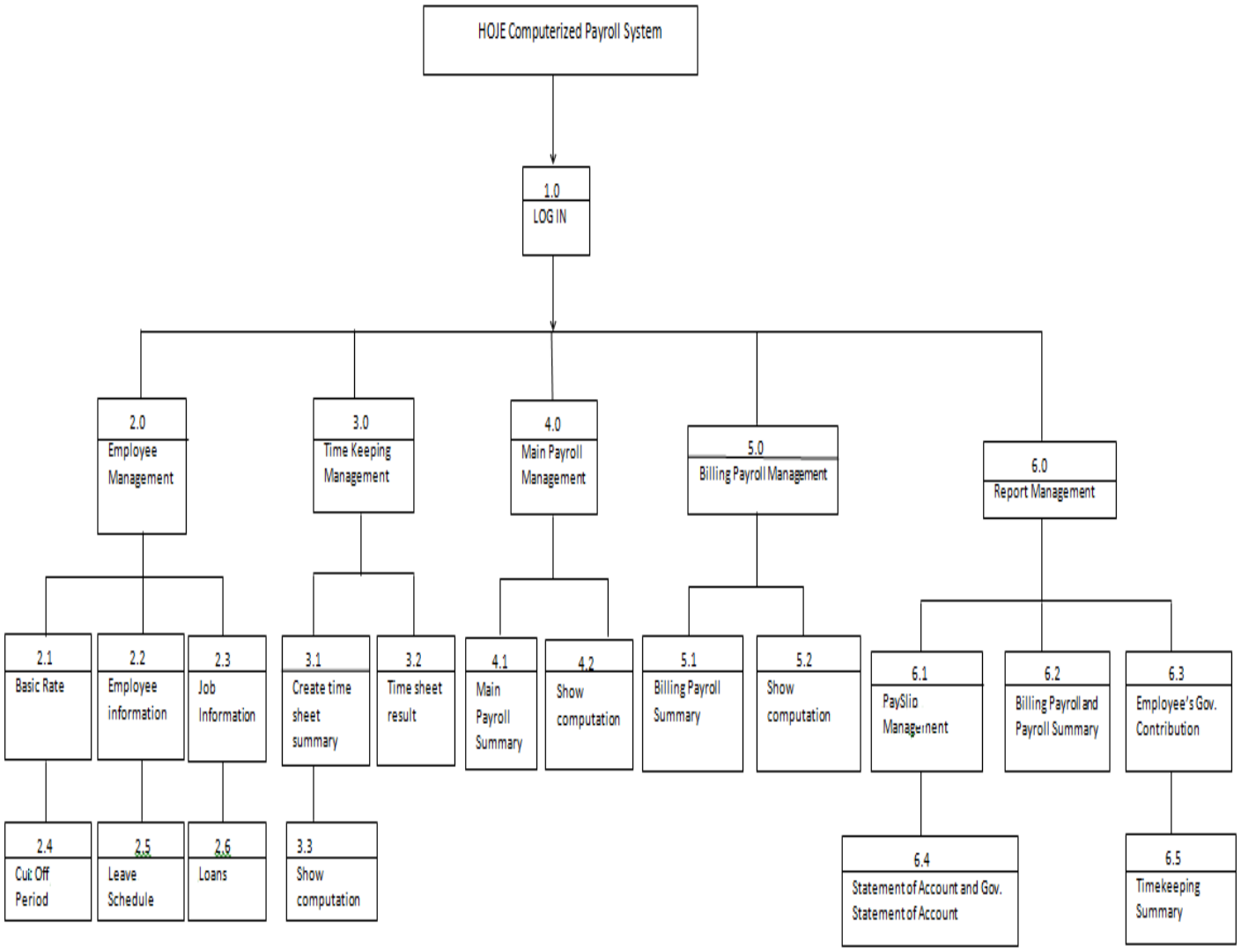


Figure 3-14 Heirarchical Input-Process-Output

Figure 3-14 shows the Heirarchical InPUT-Process-Output of te proposed system.

3.5 Entity Relationship Diagram

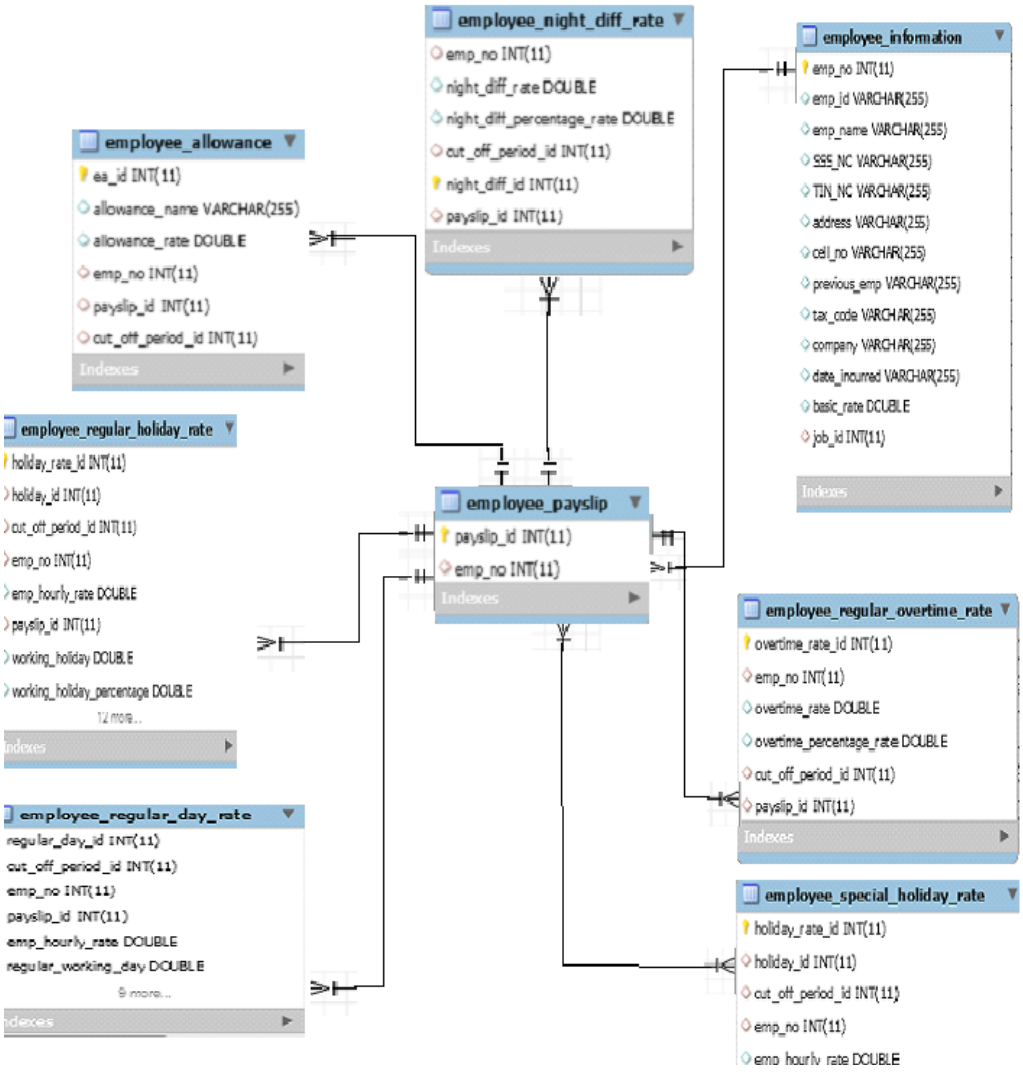


Figure 3-15 Entity Relationship Diagram (Level 0)

This figure shows the relationship among all the tables of pay slip form in employee's gross pay.

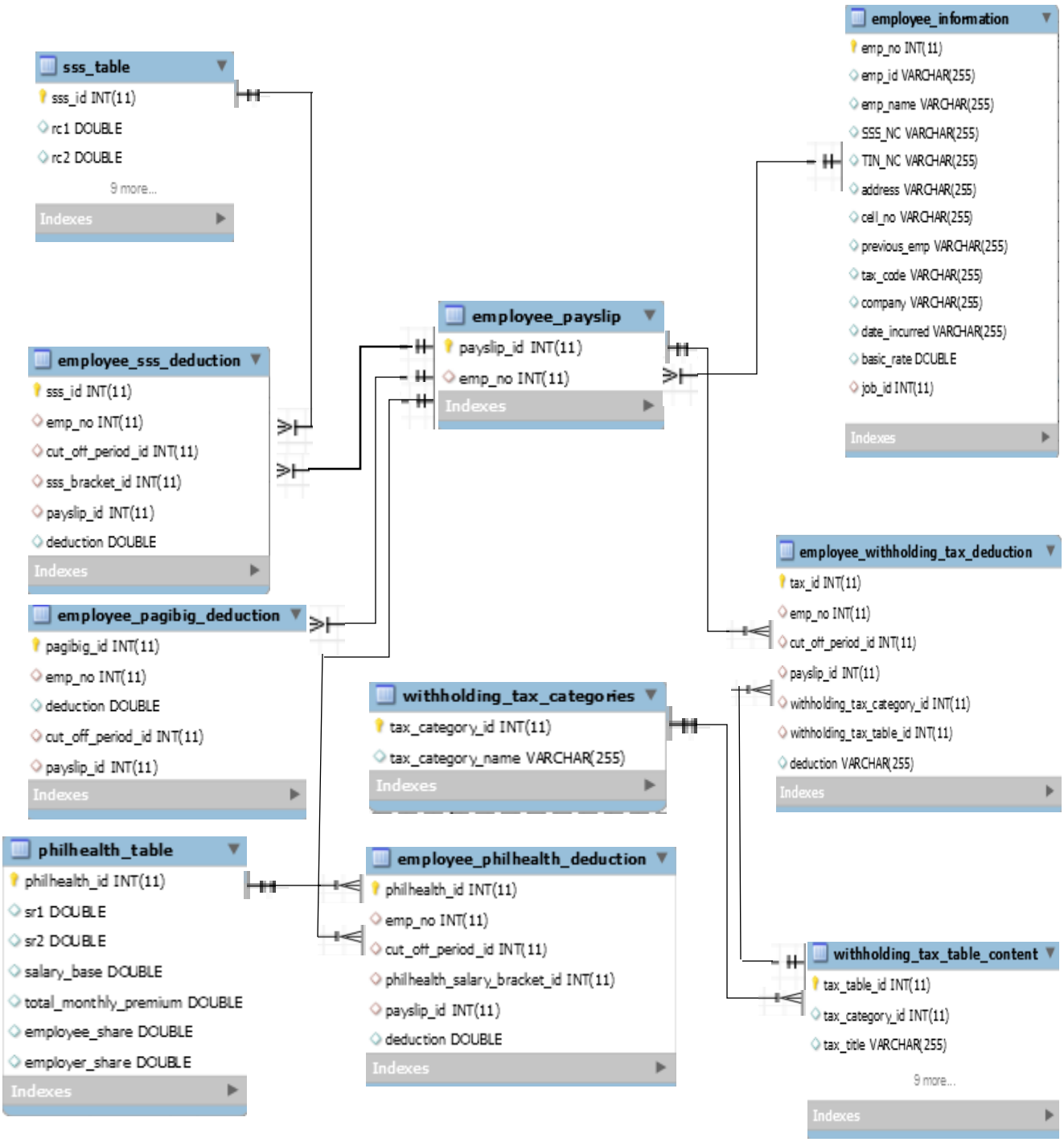


Figure 3-16 Entity Relationship Diagram (Level 1)

This figure shows the relationship among all the tables of pay slip form in employee's government deductions.

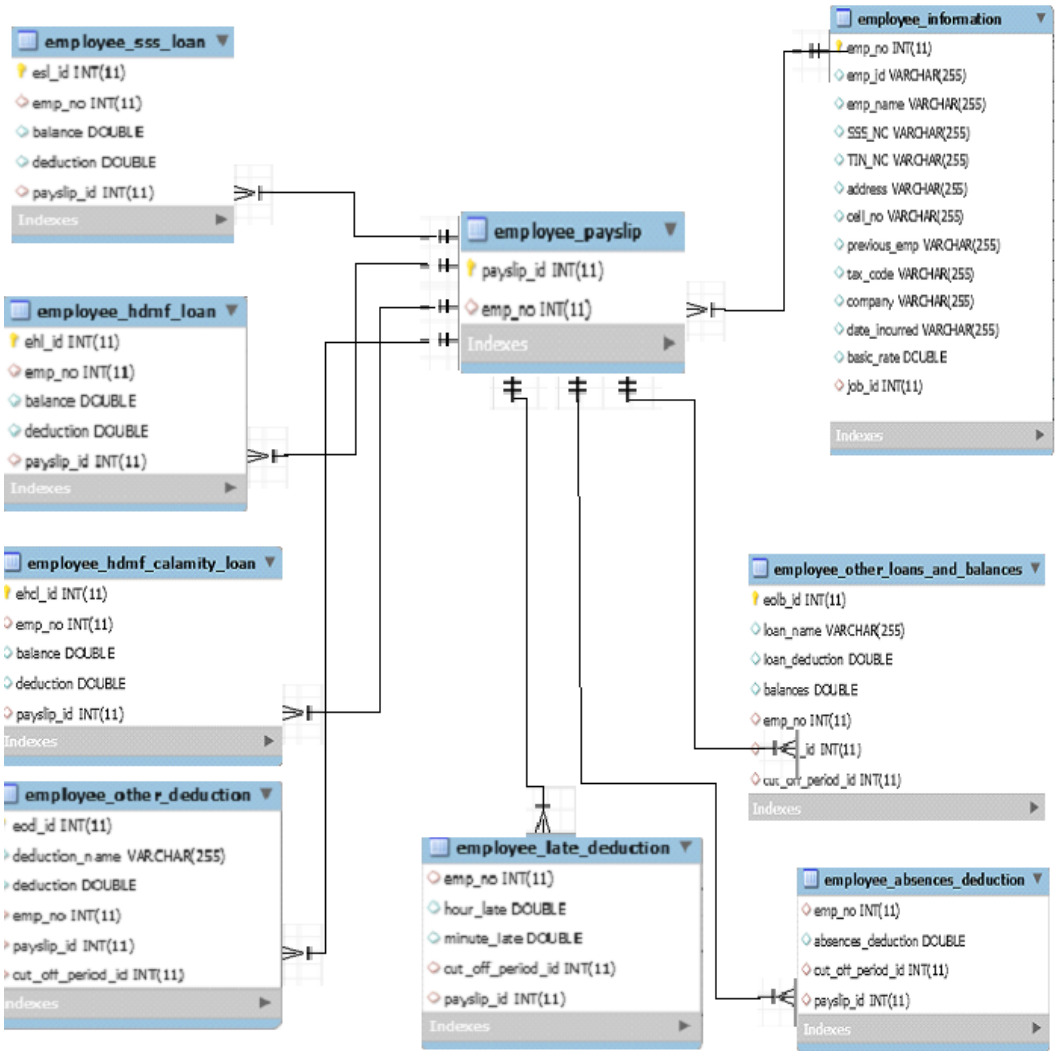


Figure 3-17 Entity Relationship Diagram (Level 2)

This figure shows the relationship among all the tables of pay slip form in employee's loan deduction and other deductions.

3.6 System Design Specification

3.6.1 Hardware (Developer and End User)

HARDWARE	DESCRIPTION
System Unit	
Processor	At least Core 2 Duo
Memory	At least 3GB RAM
Hard Disk	At least 40 GB
Monitor	Typical Monitor
Mouse	Typical Pointing Devices
Standard 128 keys	Standard 128 keys

Table 3-1 Hardware Requirements

Table 3-1 shows the minimum hardware requirements of the proposed system. Hardware requirements refer to the physical components of the computer and other peripheral that needed by the developer and end user to run the propose system efficiently.

3.6.2 Software

SOFTWARE	DESCRIPTION
Atleast windows 7 or8	Operating System (OS)
Wamp Server	Web Server
My SQL	Database Management

Table 3-2 Software Requirements for Developer

Table 3-2 shows the software requirement needed by the end user which is the Hoje.

3.7 Summary

In this Chapter, it shows all the diagrams that have been used in developing the system. It also about how the system flows in every process. And also HIPO (Hierarchical Input Process Output) diagram that how's the Accountant and Administrator can access on the payroll program. The administrator is the one who controls everything on the system. There is also a Data Flow Diagram that explains the flow of the whole software system. It explains all the flow of the system the context diagram, the level 0 and 1, the Entity Relationship Diagram that explains the relationship of the database tables.

Chapter 4 PERFORMANCE ANALYSIS

4.1 Introduction

In this chapter, the usability and function of the system will be analyzed. The developers will provide the data gathered from the relevant test, subjected to the system in order to measure the function and the usability.

The developers make a series of survey to the user, employee, and other staff of Hoje to know their reaction about the current Payroll Management of the client. The developers made some testing to verify that the objective of the system is functioning and that will evaluate how system is functioning and that will evaluate how our system functions.

4.2 Experimental

By checking the possible errors of the system and measuring its performances, the developerss conduct altered trial and errors as well as testing its technical parts and usability of the system to the users.

The following test conducted is:

- **System Evaluation**

The evaluation and function of the system is done through online base only. The developers see to it that the plan and function of the proposed system can handle foregoing and current data. The section arrange on how the system is formed and created as well as implemented to the client.

- **Unit Testing**

Unit Testing is a method by which individual units of source code are tested to determine if they are fit for use. A unit is a smallest testable part of an application. In a procedural programming a unit maybe and individual function or procedure, unit test are created by programmers. In this stage, the developers tested each module and procedure of the codes for errors and facilities exception handling.

- **Integration System**

Integration system in this phase in software testing in which individual software modules are combined and tested as group. It occurs after unit testing and before system testing. The purpose of integration testing is to verify functional, performance and reliability requirements placed on major design item. In this phase the developers merged all the modules being tested perform its functionality as an integral part of the system.

- **System Testing**

After actions were made the developers made a test run. Here. The adviser made several suggestions to make the system perform better. The same tests were performed. The overall functionality of the system meets the main objectives with some necessary adjustments, and modifications. System testing makes certain module that relates to the function of the system

No of Descriptor	**Range Scale**	**LevelofAgreement**	**Interpretation**
5	**4.21-5.00**	**StronglyAgree (SA)**	**Excellent**
4	**3.41-4.20**	**Agree (A)**	**Average**
3	**2.61-3.40**	**Fairly Agree (FA)**	**Below Average**

2	**1.81-2.60**	**Disagree (D)**	**Poor**
1	**1.00-1.80**	**Strongly Disagree (SD)**	**Very Poor**

Table 4-1 Statistical Analysis

The weighted mean of each item in the user evaluation of the proposed system was computed using the following formula:

$$\frac{WM = (5(fsa)+4\ (fa)+3(ffa)+2(fd)+1(fds))/10}{N}$$

Where: WH = weighted mean

Fsa = number of evaluator who strongly agree

Fa = number of evaluator who agree

ffa = number of evaluator who fairly agree

fd = number of evaluator who disagreed

fds = number of evaluator who strongly disagreed

N = number of evaluators

b. User Acceptance

The developers allow the client accountant of HOJE to learn the system to get their ideas about process of the system. Furthermore, survey, questions are also given to the client to know if there is anything that must comprised, changed or enhanced in the system to make if more proficient.

4.3 Result and Analysis

System/ Software Evaluation Plan

This thesis uses the ISO 9126 Quality Model as a suitable tool for evaluating the system. ISO 9126 Model develop an International Organization for Standardization and is one large group of international organization that established recognized standards appropriate across a wide range of system. The developers had a use this since it has potential to deliver useful evaluation tool to gather information and feedback from end users. The develop system evaluated through a checklist questionnaire to determine its acceptability on the ISO 9126 standard. This standard determines the quality of the software product throughsoftware quality model identifies 6 main quality characteristics, namely: Functionality, Reliability, Usability, Efficiency, Maintainability, and Portability.

Characteristics	Subcharacteristics	Definitions
Functionality	Suitability	This is the essential Functionality characteristic and refers to the appropriateness (to specification) of the functions of the software.
	Accurateness	This refers to the correctness of the functions, an ATM may provide a cash dispensing function but is the amount correct?
	Interoperability	A given software component or system does not typically function in isolation. This subcharacteristic concerns the ability of a software component to interact with other components or systems.
	Compliance	Where appropriate certain industry (or government) laws and guidelines need to be complied with, i.e. SOX. This subcharacteristic addresses the compliant capability of software.
	Security	This subcharacteristic relates to unauthorized access to the software functions.
Reliability	Maturity	This subcharacteristic concerns frequency of failure of the software.
	Fault tolerance	The ability of software to withstand (and recover) from component, or environmental, failure.
	Recoverability	Ability to bring back a failed system to full operation, including data and network connections.
Usability	Understandability	Determines the ease of which the systems functions can be understood, relates to user mental models in Human Computer Interaction methods.
	Learnability	Learning effort for different users, i.e. novice, expert, casual etc.
	Operability	Ability of the software to be easily operated by a given user in a given environment.

Efficiency	Time behavior	Characterizes response times for a given thru put, i.e. transaction rate.
	Resource behavior	Characterizes resources used, i.e. memory, cpu, disk and network usage.
Maintainability	Analyzability	Characterizes the ability to identify the root cause of a failure within the software.
	Changeability	Characterizes the amount of effort to change a system.
	Stability	Characterizes the sensitivity to change of a given system that is the negative impact that may be caused by system changes.
	Testability	Characterizes the effort needed to verify (test) a system change.
Portability	Adaptability	Characterizes the ability of the system to change to new specifications or operating environments.
	Installability	Characterizes the effort required to install the software.
	Conformance	Similar to compliance for functionality, but this characteristic relates to portability. One example would be Open SQL conformance which relates to portability of database used.
	Replaceability	Characterizes the *plug and play* aspect of software components, that is how easy is it to exchange a given software component within a specified environment.

Table 4-2 ISO 9126 Characteristics and Sub-characteristics ISO 1991:Abram 2003)

Based on the experiment and assessment of the observers, the developers conduct revisions and modifications of the system.

The developers used a qualitative research approach in collecting and investigating from the client. The developers used the interview techniques to gather standards and information to make the proposed system more efficient, feasible and reliable. In collecting statistical data the developers use closed-ended questionnaire. The survey basis is the system is achieve and acceptable by the client. 1 administrator and 9 persons with background in IT and Accountancy.

By calculating the result, the proponent used the computation of General Average with the following scale :

5 Strongly agree (SA) 4 Agree (A)

3 Uncertain (U) 2 Disagree (D)

1 Strongly Disagree (SD)

Below is a sample of survey questionnaire for the system evaluation.

Software Acceptability and Features	Rating				
	5	4	3	2	1
A. Functionality of the system					
1. The software performed the task required.					
2. The software presented the results as expected.					
3. The software interacted with another system.					
4. The software is compliant with standards.					
5. The software provides security for unauthorized access.					
B. Reliability of the system					
1. The faults of the software can be eliminated.					
2. The software capable of handling errors.					
3. The software resumed working and restore data after failure.					
C. Usability of the System					
1. The software can be understand/comprehend by the users easily					
2. The software is learned easily.					
3. The software can be used /operated with less effort.					
4. The software interface looks good.					
D. Efficiency of the system					
1. The software responded to the user quickly.					
2. The software utilized resources efficiently.					

Figure 4-1 System Evaluation

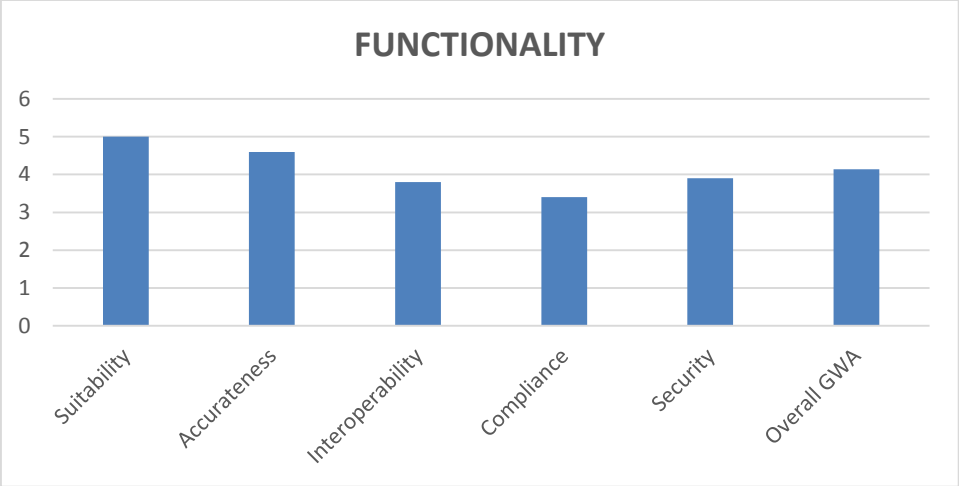


Figure 4-2 Chart for Functionality Evaluation

Figure 4-2 chart the average results on the functionality of the system. The sustainability, accuratensess and security and got the highest weighted mean 5 which is equivalent to strongly agree. The compliance criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system is 4.14 which men the system was evaluated by positively by the respondents.

Reliability Evaluation

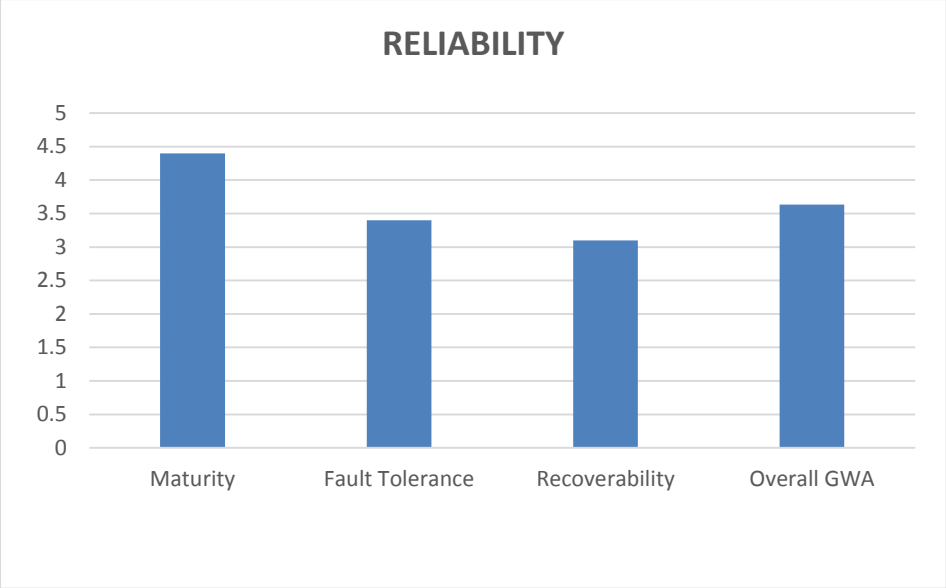


Figure 4-3 Chart for Reliability Evaluation

Figure 4-3 chart the average results on the reliability of the system. The maturity, got the highest weighted mean 4.4 which is equivalent to agree. The recoverability criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system is 3.63 which men the system was evaluated by positively by the respondents.

Usability Evaluation

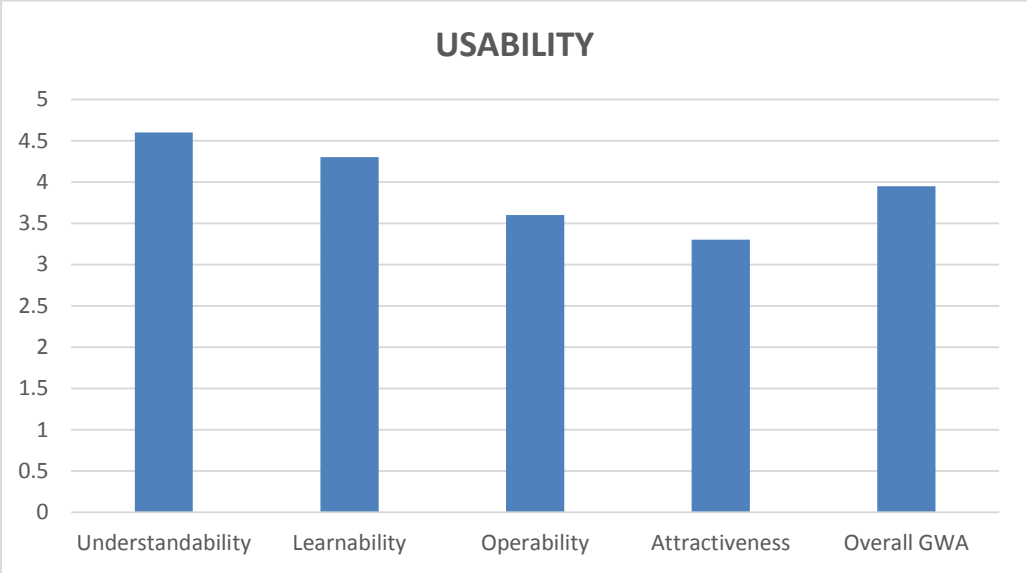


Figure 4.4 Chart for Usability Evaluation

Figure 4-4 chart the average results on the usability of the system. The understandability got the highest weighted mean 4.6 which is equivalent to strongly agree. The attractiveness criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system is 3.95which men the system was evaluated by positively by the respondents.

Efficiency Evaluation

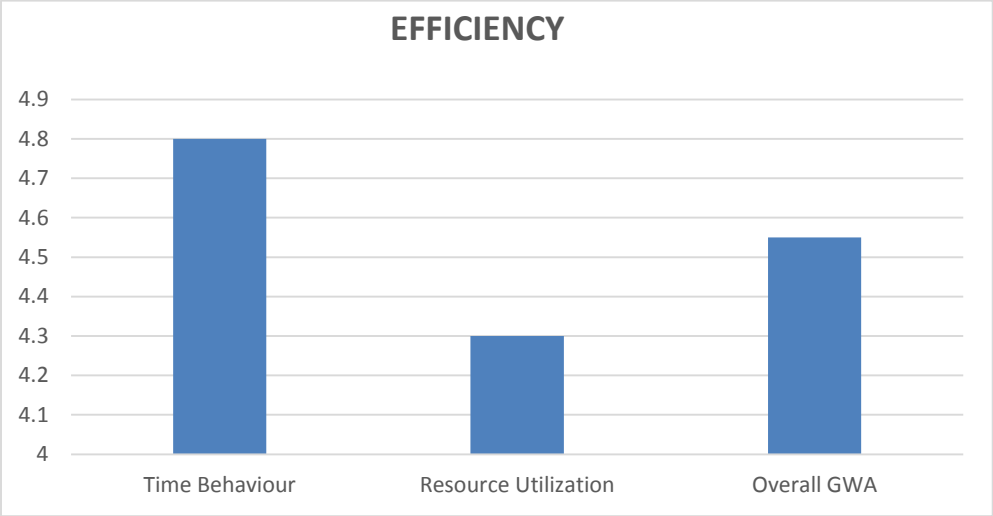


Figure 4-5 Chart for Efficiency Evaluation

Figure 4-5 chart the average results on the efficiency of the system. Time behaviour, got the highest weighted mean 4.8 which is equivalent to strongly agree. The resource utilization criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system 4.55which men the system was evaluated by positively by the respondents.

Maintainability Evaluation

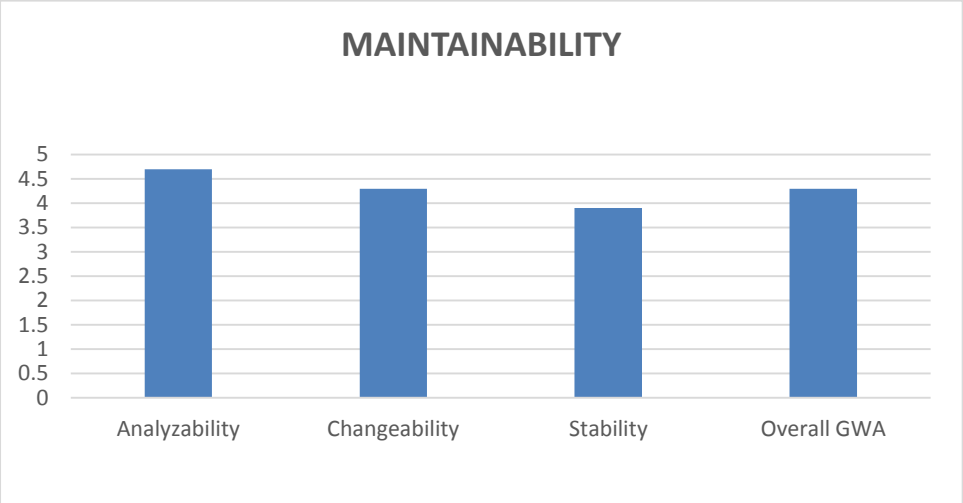


Figure 4-6 Chart for Maintainability Evaluation

Figure 4-6 chart the average results on the maintainability of the system. Analyzability got the highest weighted mean 4.7 which is equivalent to strongly agree. The stability 3.9 criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system is 4.3 which men the system was evaluated by positively by the respondents,

Portability Evaluation

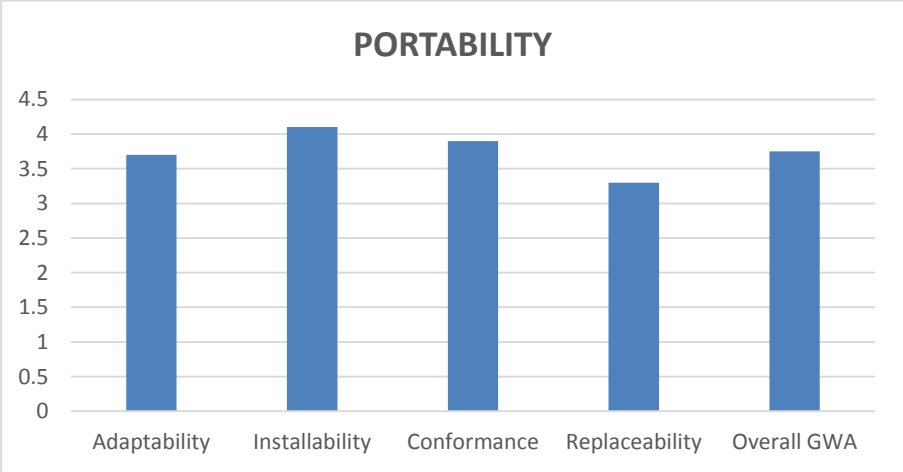


Figure 4-7 Chart for Portability Evaluation

Figure 4-7 chart the average results on the portability of the system. Installability got the highest weighted mean 4.1 which is equivalent to strongly agree. The replaceability 3.3 criterion is the lowest is strongly disagreeing. The result shows that the average weighted mean regarding the functionality of the system is 3.75 which men the system was evaluated by positively by the respondents.

Summary Evaluation

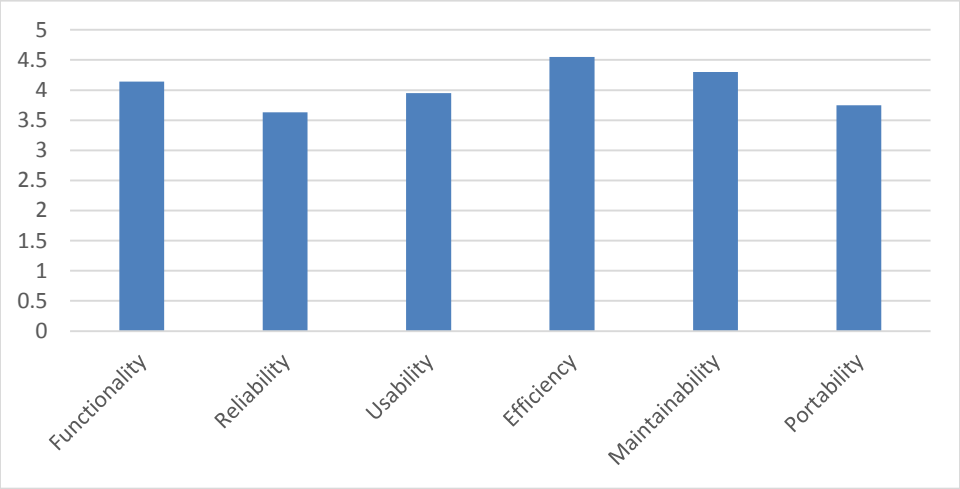


Figure 4-8 Chart for Summary Evaluation

4.4 Summary

This chapter has undertaken to illustrate the system where it can handle and secure the data in organized method and presented a series of test and evaluation of the system. The different concept and ideas that the developers accepted recognized a system where it would not only focus on the developers appreciative never the less to the demands and requirements of the client to became the aim for developing the system. The developers certify the functionality and design for the system.

Chapter 5 CONCLUSION

Hoje still uses common ways of handling payroll transaction and information. In their payroll transaction, the client can only put the payroll information in a sheet in Microsoft excel and compute it there. And in terms of saving files they will saved it in to the folder and if the files are too many. The client is always experiencing problems finding the files they need, especially with t payroll transaction.

The developers decided to create a java based system could improve the payroll transaction of the client. The system will give easier and most convenient way of managing payroll transaction.

With the help of programming languages like Java Programming Environment – NetBeans and MySQL as the database the developers accomplished the system. The developers also conducted experiment to determine the system if it's accurately computing results if all of its functionality is working properly. The developers therefore conclude that developing a payroll system will make a good improvement and will help the Home of Job Enterprises (HOJE) to make the transactions easier.

5.1 RECOMMENDATION

Computerized Payroll System is designed for Home of Job Enterprise (HOJE). The purpose of the system is to give a convenient way of preventing errors of computing payroll, and fasting the payroll transaction. The future researchers may view our proposed system to use as their reference.

5.2 SUMMARY

The developers achieved its objectives in conducting this study. The client is satisfied because all the expectations have been meet. HOJE can now serve their clients with the help of Computerized Payroll System. The significances of this study are to work as a guide in improving or enhancing the new version of the system.

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APPENDIX A

(Letter of Acceptance)

APPENDIX B

(Transcript of Interview)

APPENDIX C

(Screenshots of Program)

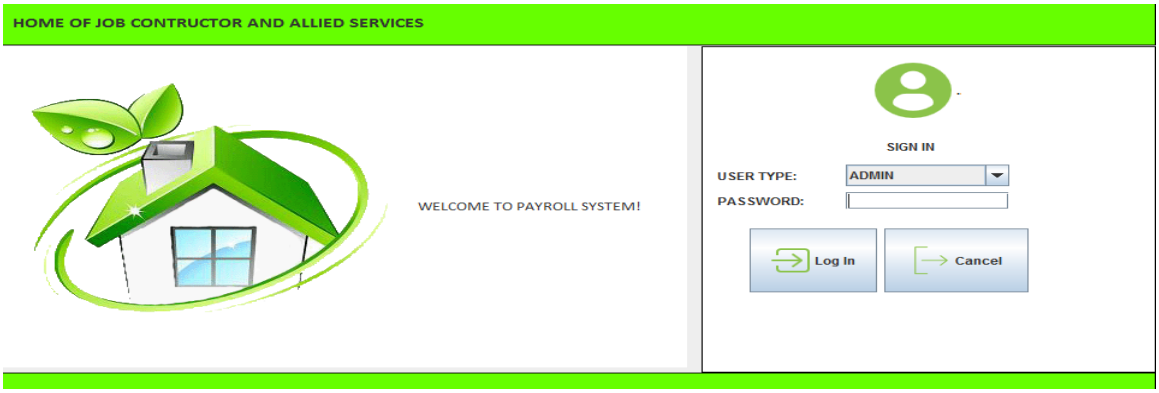


Figure C-1.Login Module



Figure C-2. Login Failed Module

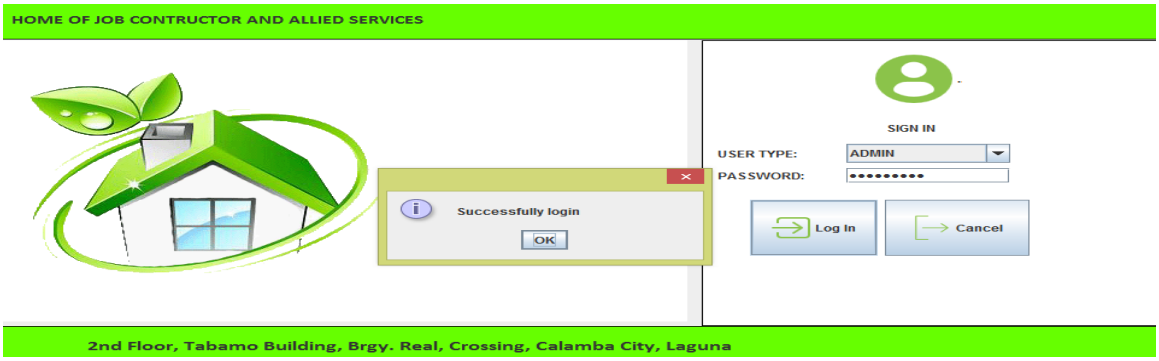


Figure C-3 Successful Login Module

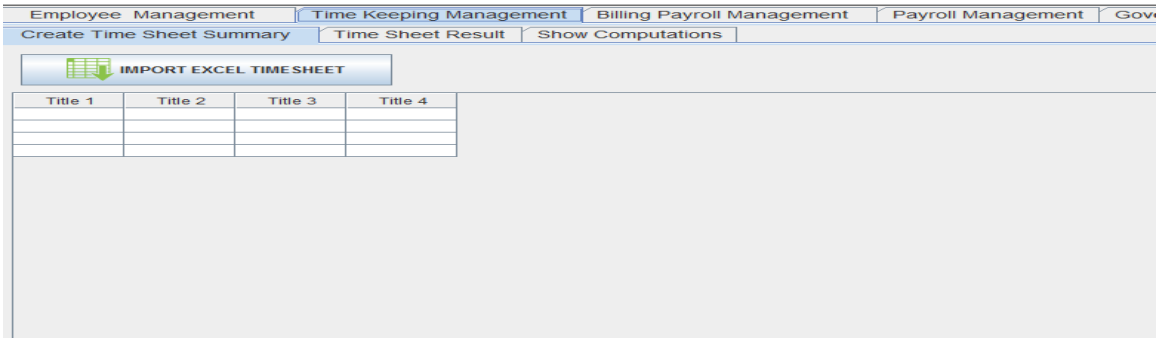


Figure C-4 Time Keeping Management Module

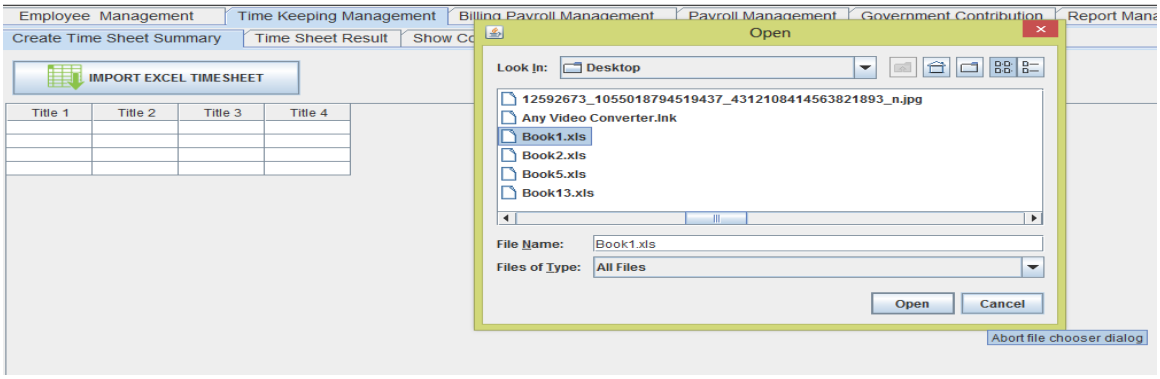


Figure C-5 Create Timesheet Module

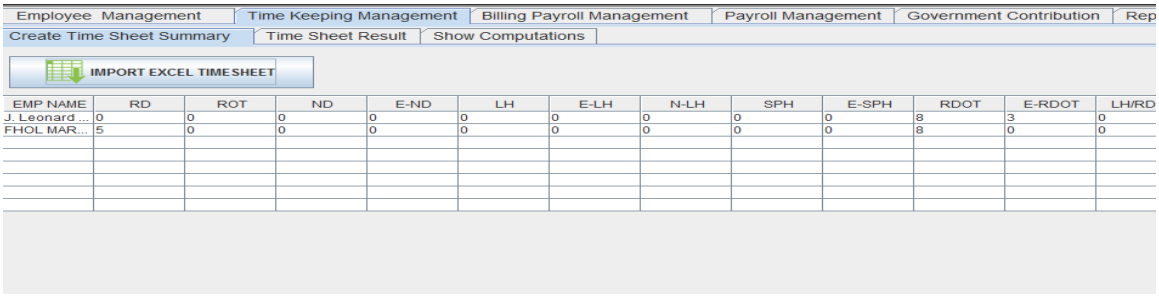
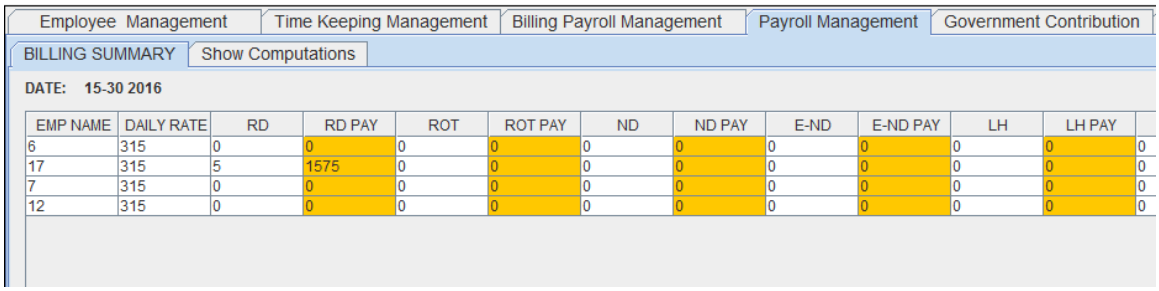
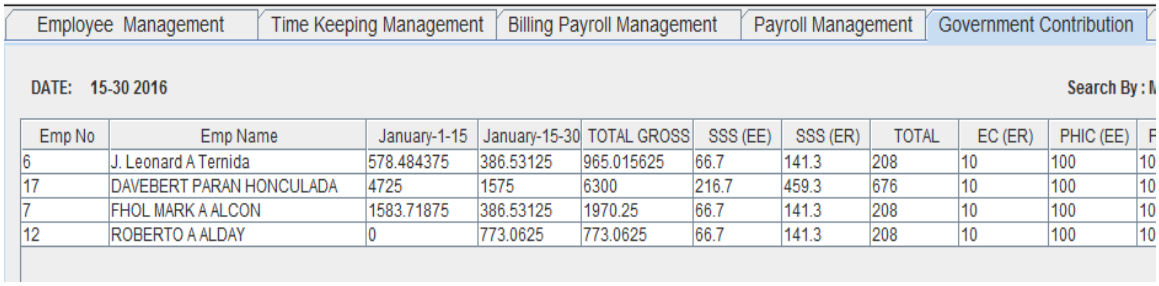


Figure C-5 Create Timesheet Summary Module



EMP NAME	DAILY RATE	RD	RD PAY	ROT	ROT PAY	ND	ND PAY	E-ND	E-ND PAY	LH	LH PAY	
6	315	0	0	0	0	0	0	0	0	0	0	0
17	315	5	1575	0	0	0	0	0	0	0	0	0
7	315	0	0	0	0	0	0	0	0	0	0	0
12	315	0	0	0	0	0	0	0	0	0	0	0

Figure C-6 Payroll Management Module



Emp No	Emp Name	January-1-15	January-15-30	TOTAL GROSS	SSS (EE)	SSS (ER)	TOTAL	EC (ER)	PHIC (EE)	P
6	J. Leonard A Ternida	578.484375	386.53125	965.015625	66.7	141.3	208	10	100	10
17	DAVEBERT PARAN HONCULADA	4725	1575	6300	216.7	459.3	676	10	100	10
7	FHOL MARK A ALCON	1583.71875	386.53125	1970.25	66.7	141.3	208	10	100	10
12	ROBERTO A ALDAY	0	773.0625	773.0625	66.7	141.3	208	10	100	10

Figure C-7 Government Contribution Module

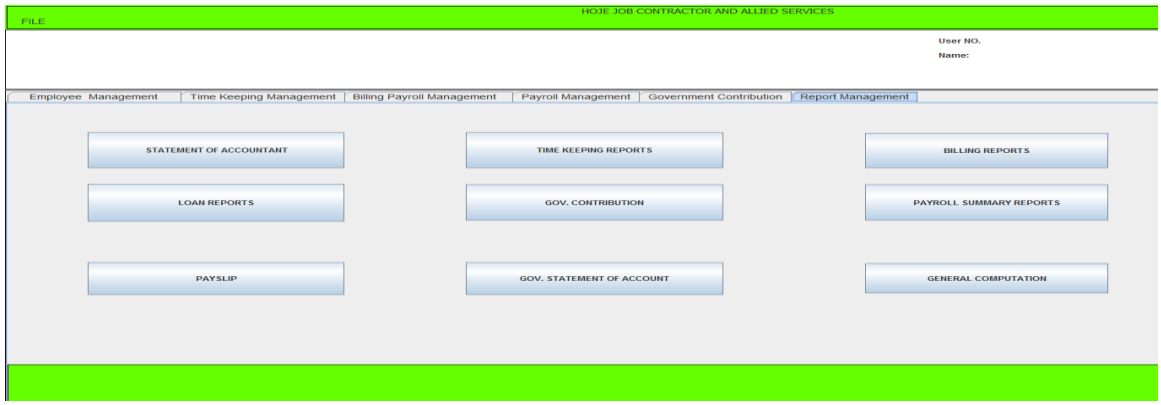


Figure C-8 Report Management Module

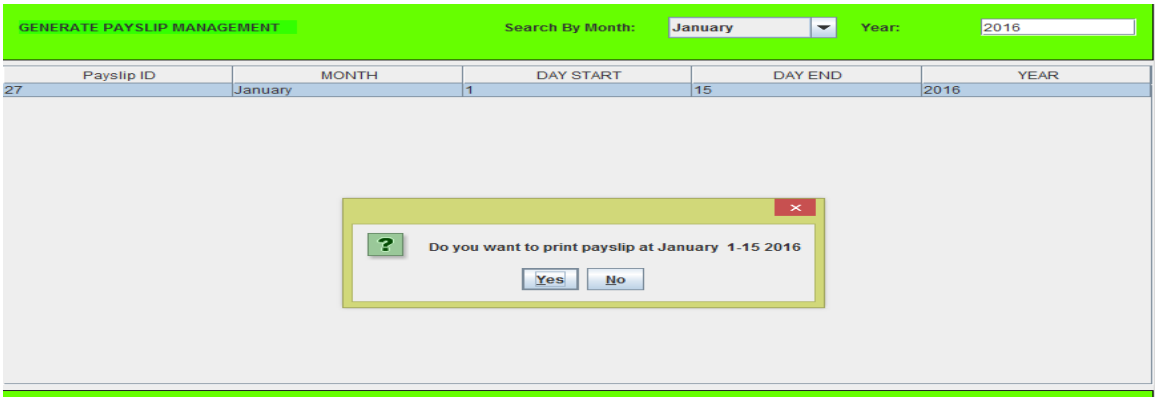


Figure C-9 Generate Payslip Management Module

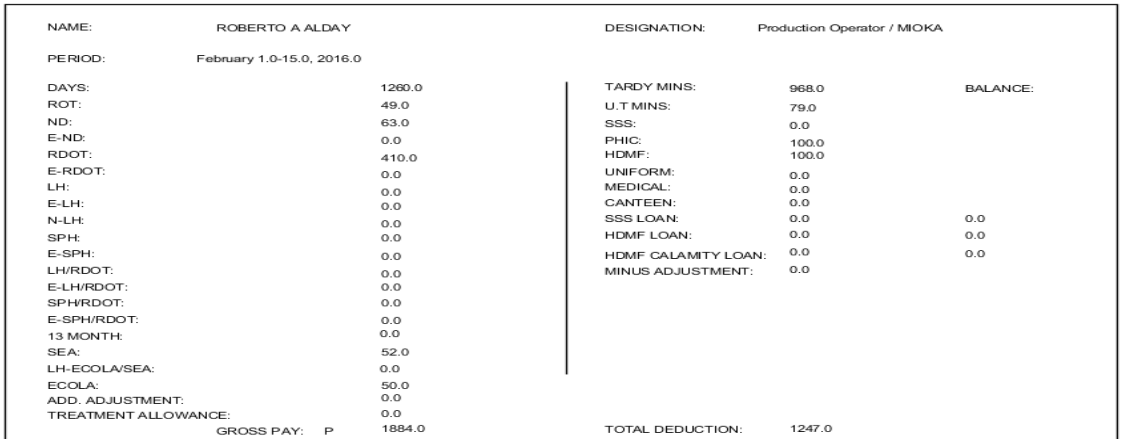


Figure C-10 Sample Payslip Module

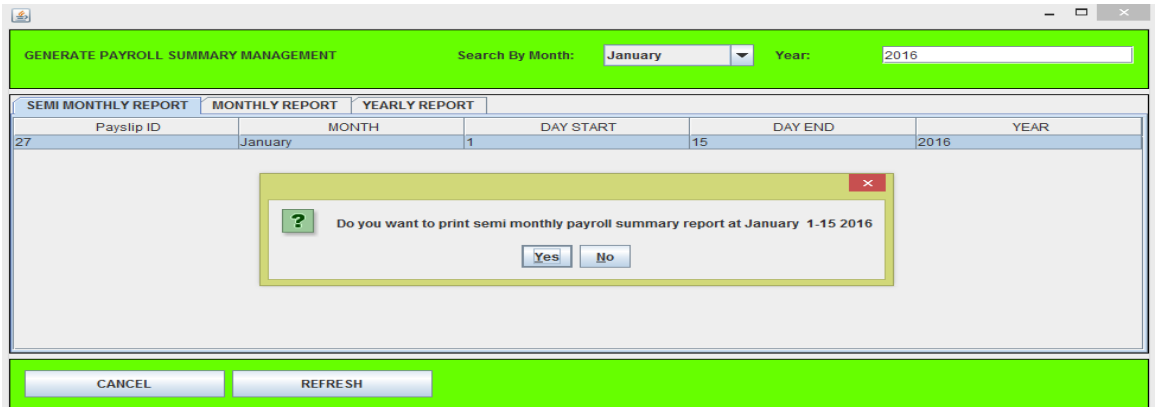
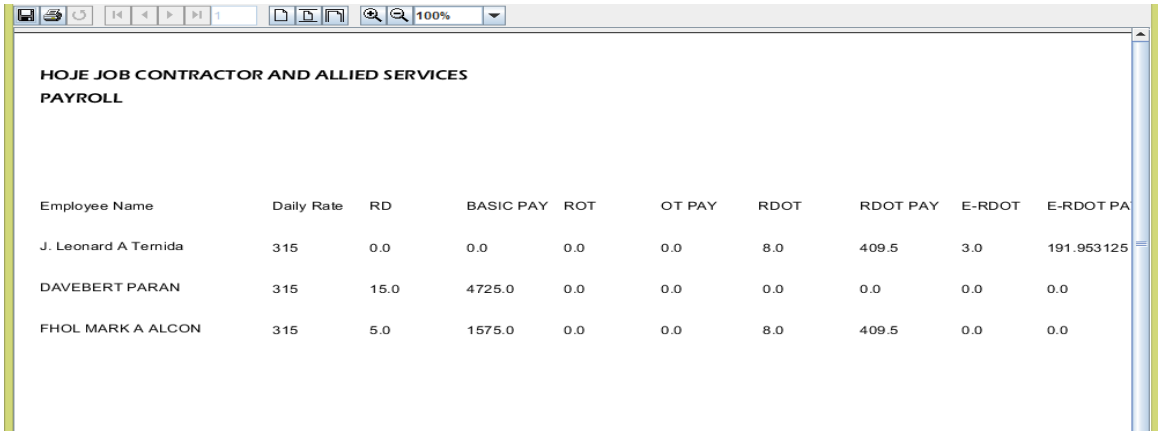


Figure C-11 Generate Payroll Management Module



HOJE JOB CONTRACTOR AND ALLIED SERVICES
PAYROLL

Employee Name	Daily Rate	RD	BASIC PAY	ROT	OT PAY	RDOT	RDOT PAY	E-RDOT	E-RDOT PA
J. Leonard A Ternida	315	0.0	0.0	0.0	0.0	8.0	409.5	3.0	191.953125
DAVEBERT PARAN	315	15.0	4725.0	0.0	0.0	0.0	0.0	0.0	0.0
FHOL MARK A ALCON	315	5.0	1575.0	0.0	0.0	8.0	409.5	0.0	0.0

Figure C-12 Payroll Module

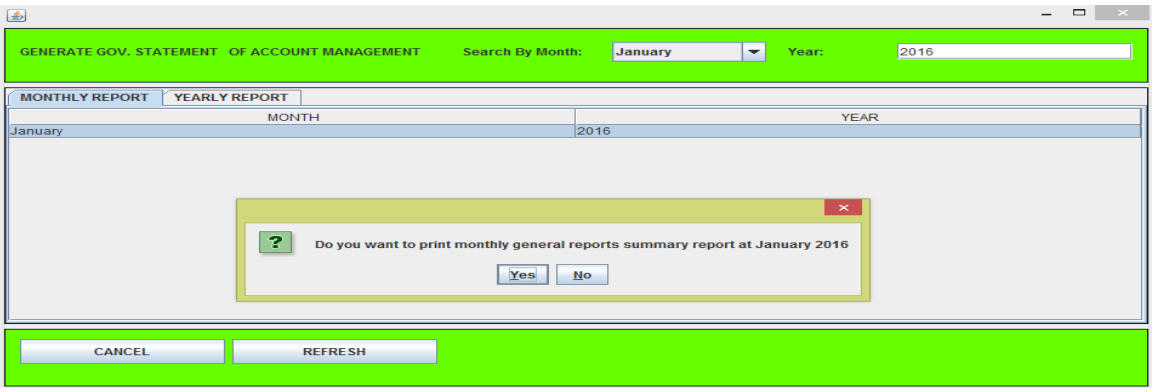


Figure C-13 Monthly Report Module

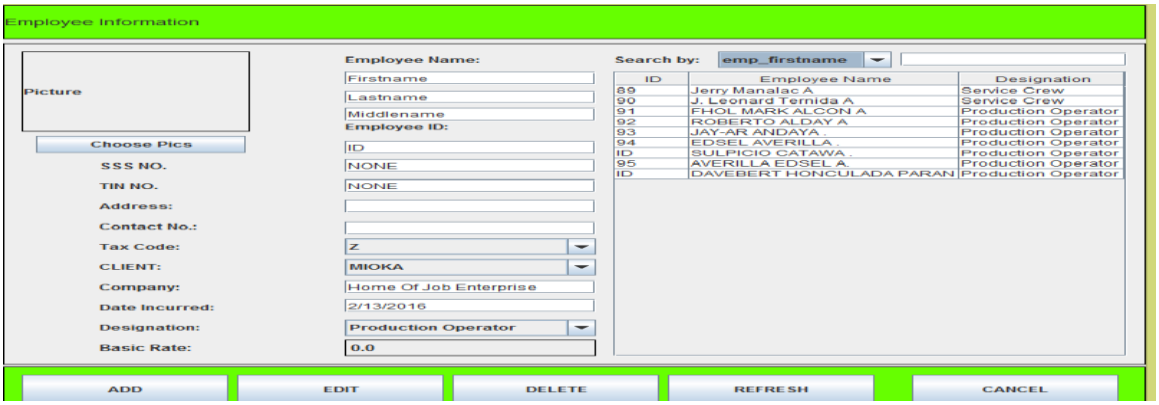


Figure C-14 Employee Information Module

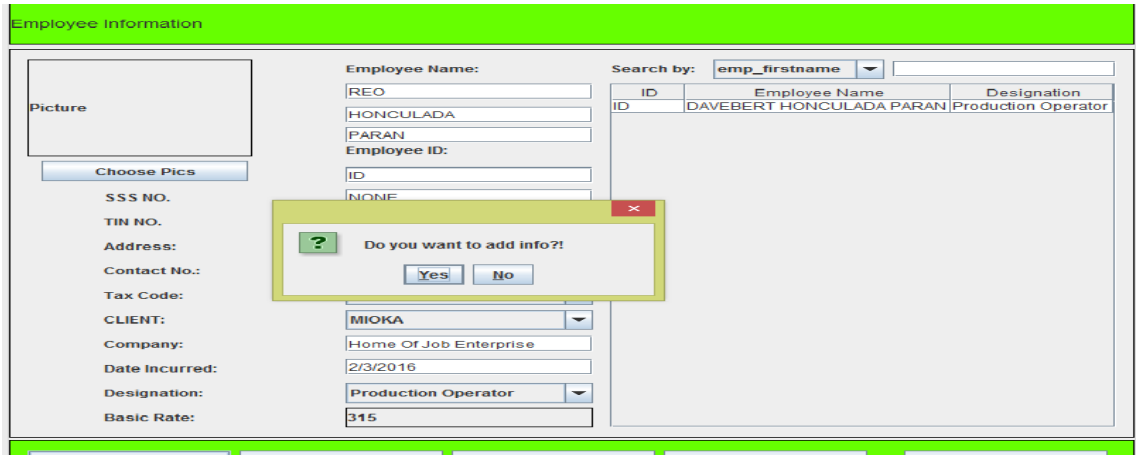


Figure C-15 Employee Information Module



Figure C-16 Contribution Table Module

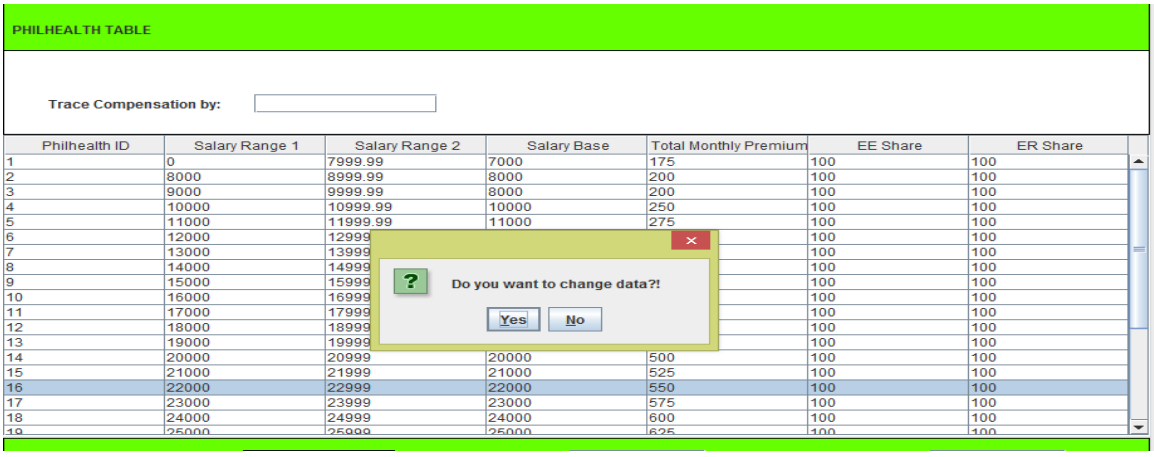


Figure C-17 Philhealth Table Module

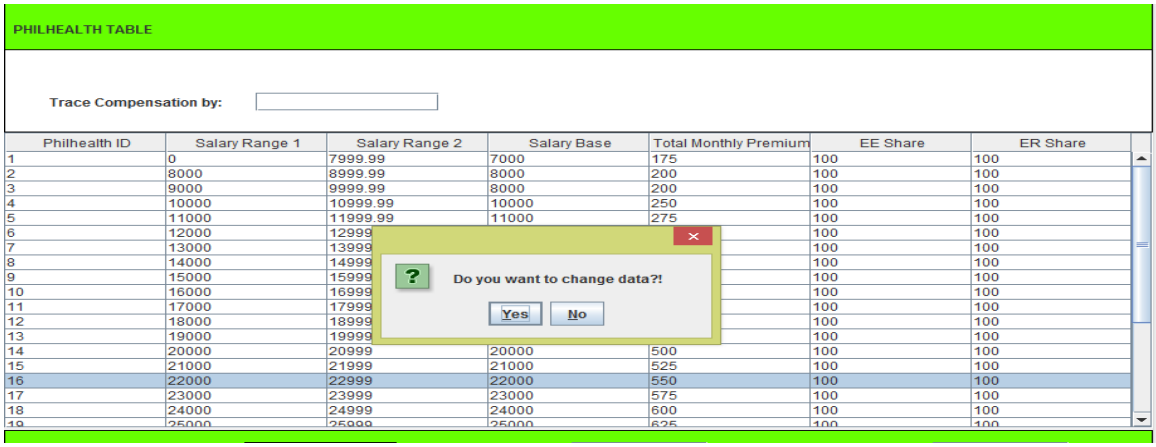


Figure C-18 Philhealth Contribution Table Module

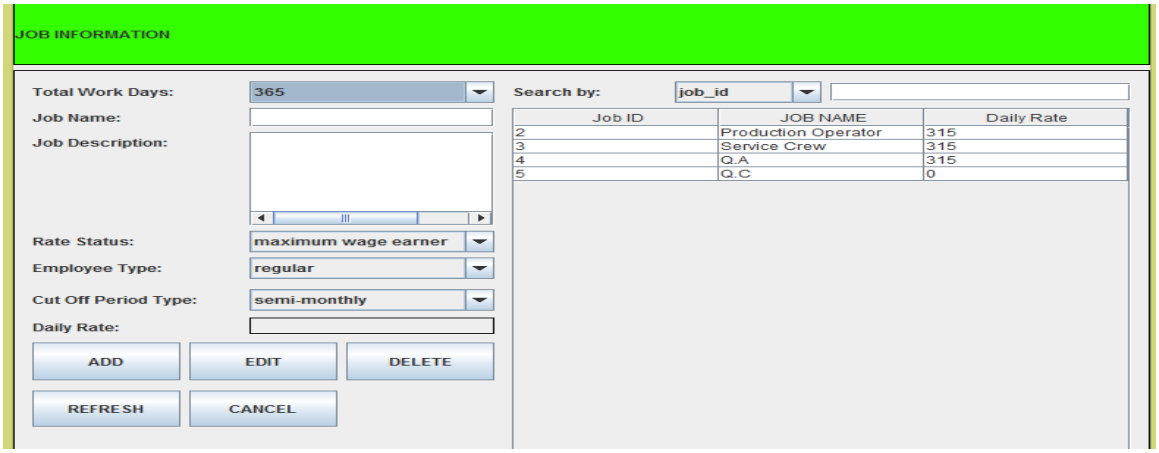


Figure C-19 Job Information Module

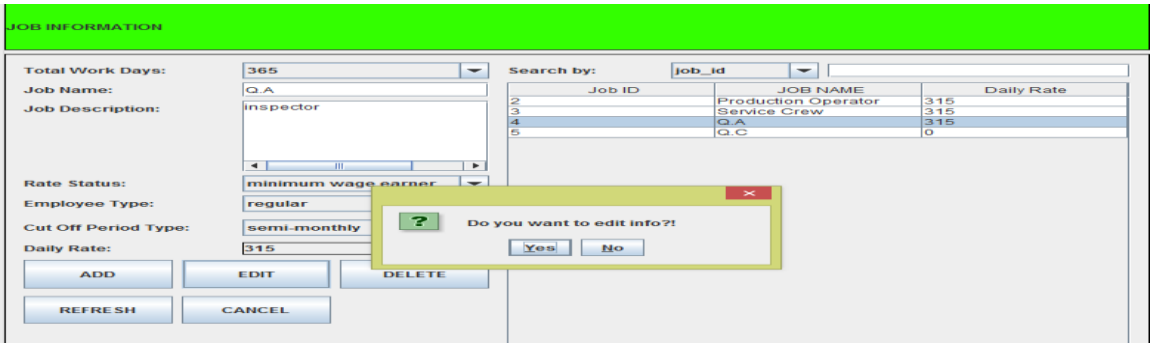


Figure C-20 Job Information Module

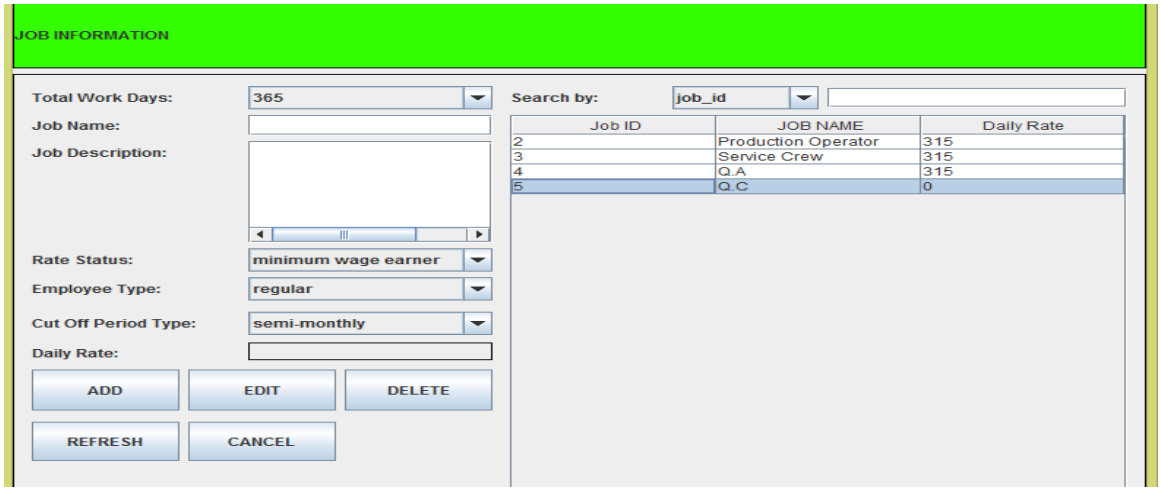


Figure C-21 Job Information Module

APPENDIX D

(Journal)

Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	

Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	
Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	

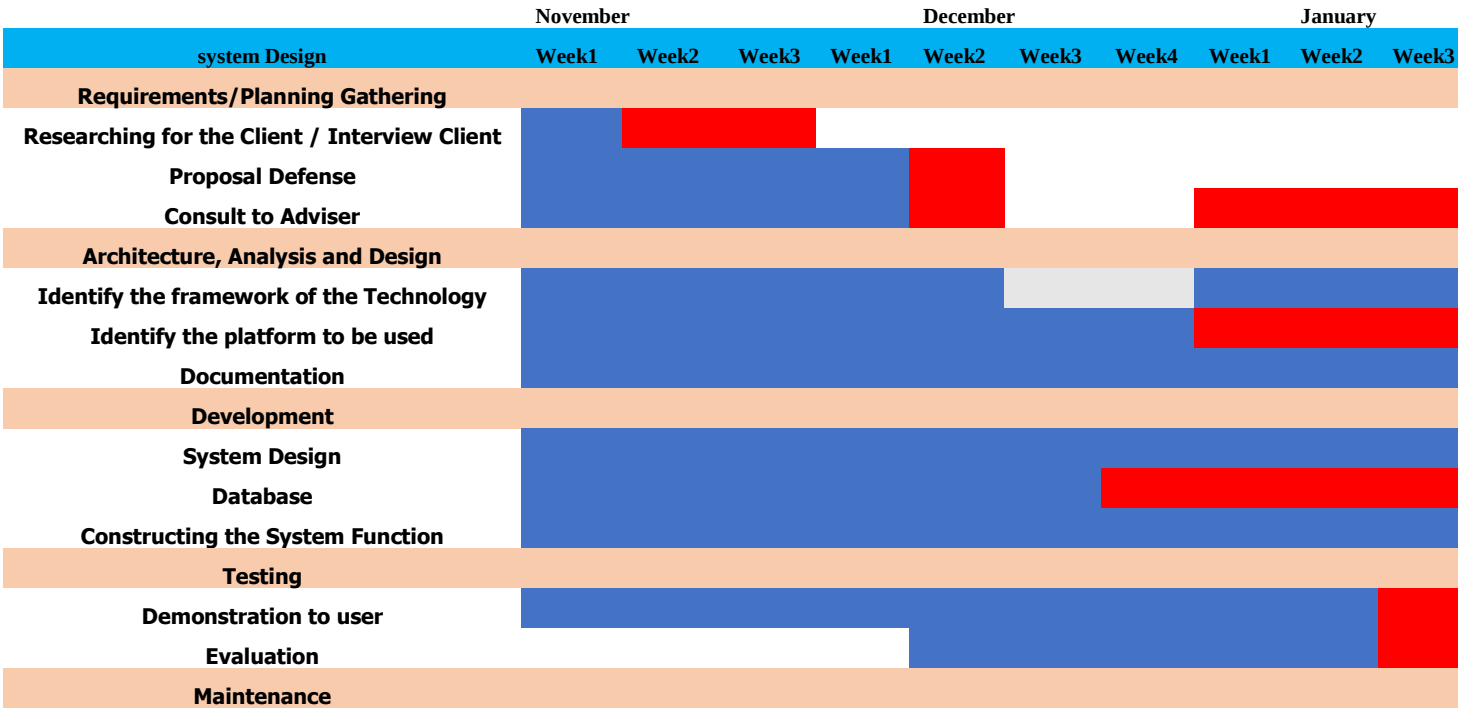
Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	

Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	

Journal	
Date : **Time :**	
Meeting :	
Members : Alcaraz, Jasmine M.	
Dela Cruz, Judy Ann R.	
Ternida,, John Leonard A.	
Adviser : Sir Jerald Sanchez	

APPENDIX E

(Calendar of Activities)



Finish

On going

Figure E-1-Gantt Chart

APPENDIX F

(Program Listing)

APPENDIX G

(Curriculum Vitae)

JOHN LEONARD A. TERNIDA

Brgy. Banadero Calamba City

09217235332

leonardternida_11@yahoo.com



Objectives:

- I am seeking employment with a company where I can use my talents and skills to grow and expand the company.
- I want to succeed in stimulating and challenging environment, building the success of the company while I experience advancement opportunities.
- I want to excel in this field with hard work, perseverance and dedication.

Qualification:

Experiencing computer programming of java and database development

in 4 years course of Information Technology in STI CALAMBA.

Programming Languages:

Java Netbeans and Visual Basic 6.0

Database Development:

mysql database

Educational Background:

Tertiary Education:

STI College Calamba

National Hi-way, Brgy.Uno 4027 Calamba, Laguna, Philippines

Bachelor of Science in Information Technology

2012-present

Secondary Education:

Calamba National High School

March 29, 2012- Graduated

Primary Education:

Calamba Elementary School

April 2, 2008- Graduated

Other informations:

Activities and skills:

Programming, drawing and guitar

Achievements:

Junior Police Award